



Statistics & Data Analysis Quickstart

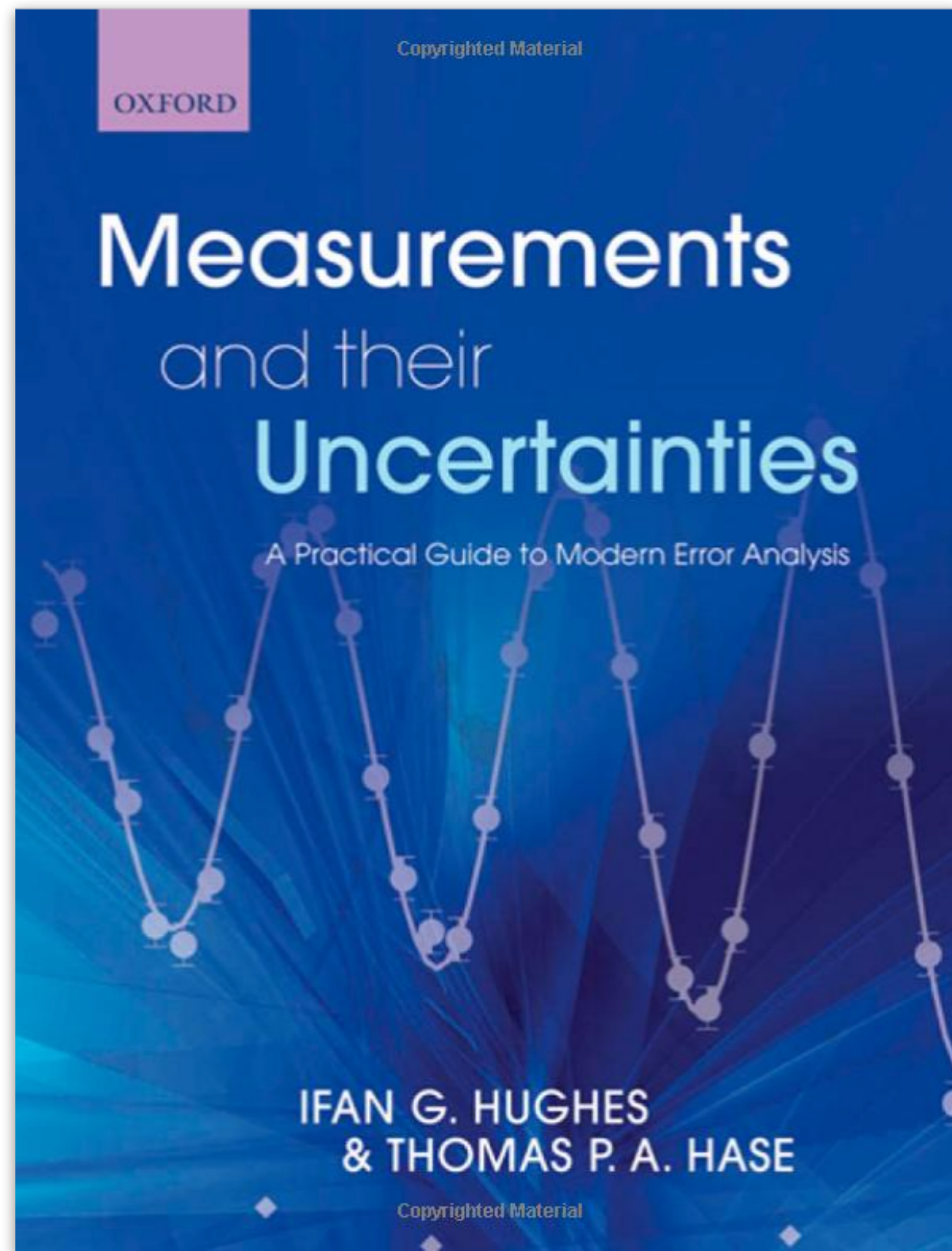
Stage 3 Advanced Labs

Prof. John Quinn

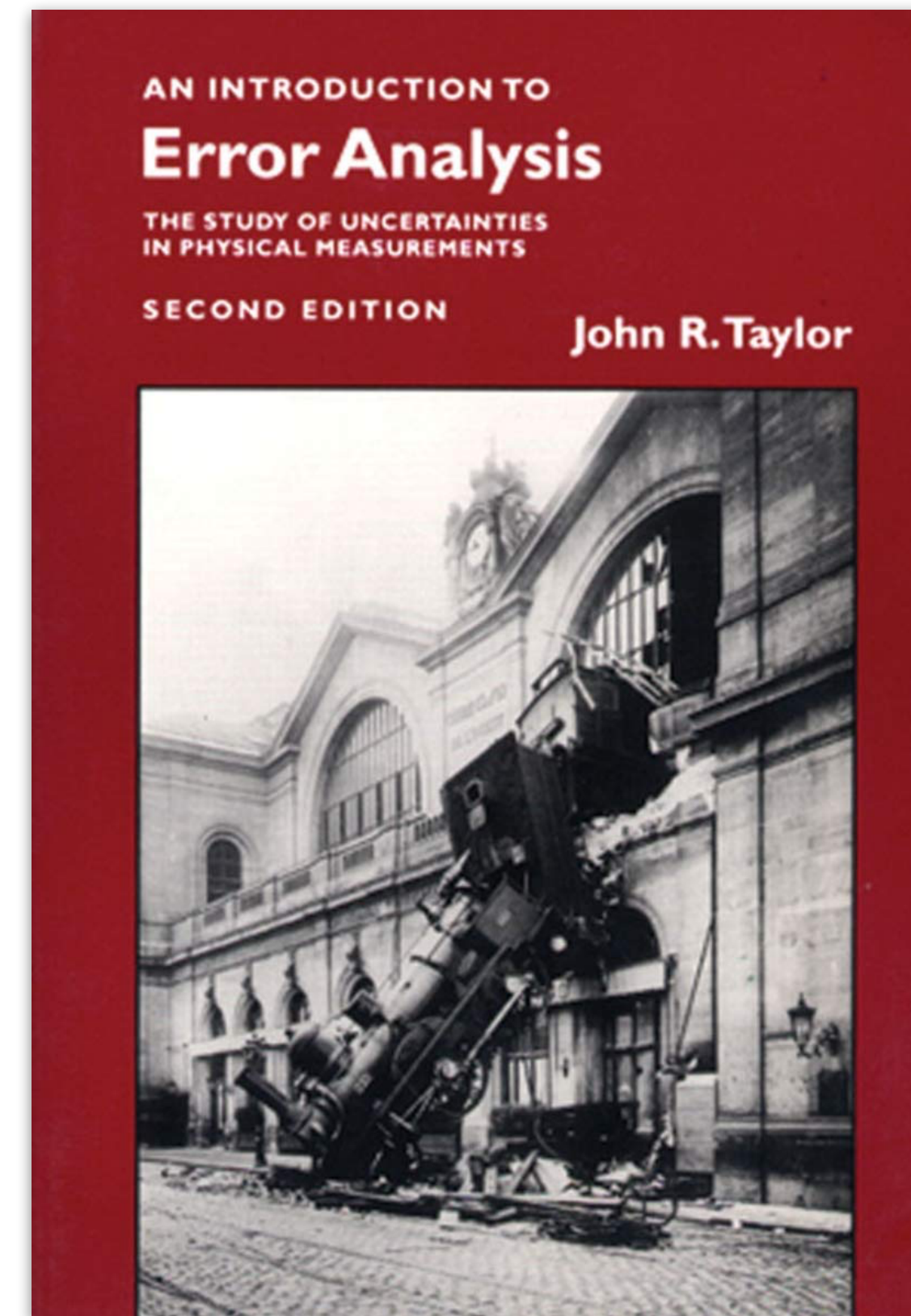
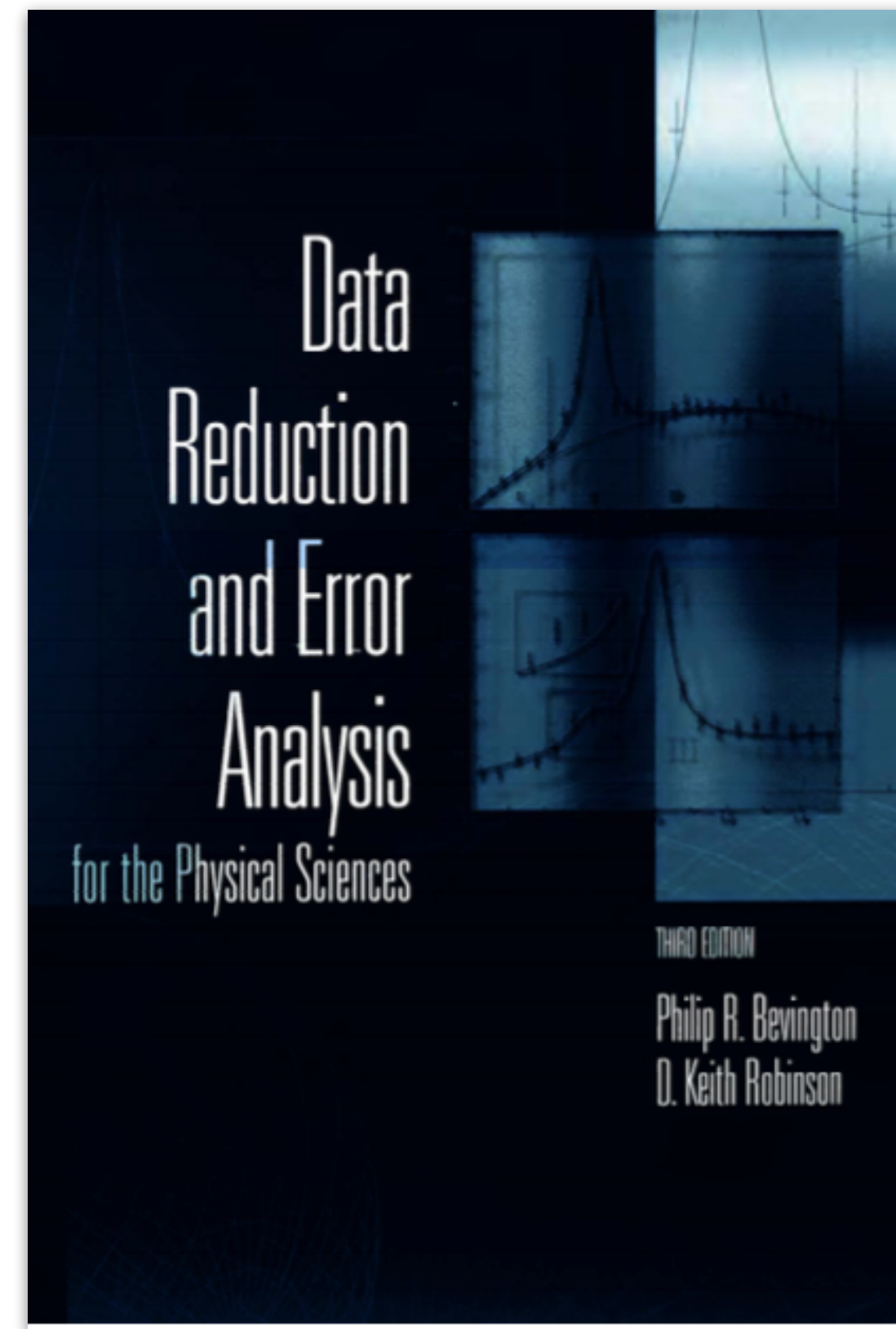
Data Analysis in Advanced Labs

- All analysis to be done in **Python** in Jupyter Notebook / Lab.
 - i.e. **no Excel!**
 - include notebook printout (PDF) as appendix to main report and include graphs, fits, calculation results (i.e. do not just give the code, do not give screenshots of code!)
- `Scipy.optimize.curve_fit()` is the fitting package to be used.
- **Error analysis** should be attempted and sources of error discussed.
 - Please be explicit and clear about your error analysis.
- Make sure to have an "**Analysis**" section in your report (if relevant) where the analysis steps are described without code:
 - The "**Results**" section follows and can include summary tables, graphs etc.
- Please quote all numbers with appropriate precision.

Some Recommended Books



UCD Online:
<http://lib.myilibrary.com/Open.aspx?id=273234#>



<https://www.iso.org/standard/50461.html>





Lockdown Tutorials

- Slides and recordings at: https://physicslabs.ucd.ie/docs/data_analysis/

The screenshot shows a web browser window displaying the UCD Physics Labs website. The browser's address bar shows the URL `physicslabs.ucd.ie/documentation/data_analysis/`. The website has a blue header with the UCD Physics Labs logo and a search bar. The main content area is titled "Data Analysis" and contains several sections: "Introduction", "Measurement Probability Distributions:", "Propagation of Errors and the Variance-Covariance Matrix:", "Curve Fitting:", and "(Computer) Matrix Propagaion of Errors:". Each section lists links to slides and zoom recordings. A left sidebar contains navigation links for "UCD Physics Labs", "Information", "Documentation", and "Report Writing". A right sidebar contains a "Table of contents" for the current page.

UCD Physics Labs

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Beech Hill

Stage 3

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Dignity and Respect

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Documentation

Python

DACQ

Data Analysis

Report Writing

Data Analysis

Introduction

Here are links to the presentations as part of the Wednesday Data Analysis Tutorials for the Advanced Lab. given in 2020 in place of student seminars.

Measurement Probability Distributions:

- [slides.pdf](#)
- Zoom Recordings: [2020-10-14.mp4](#), [2020-10-21.mp4](#)

Propagation of Errors and the Variance-Covariance Matrix:

- [slides.pdf](#)
- Zoom Recordings: [2020-10-28.mp4](#)

Curve Fitting:

- [slides.pdf](#)
- Zoom Recordings: [2020-11-04.mp4](#)

(Computer) Matrix Propagaion of Errors:

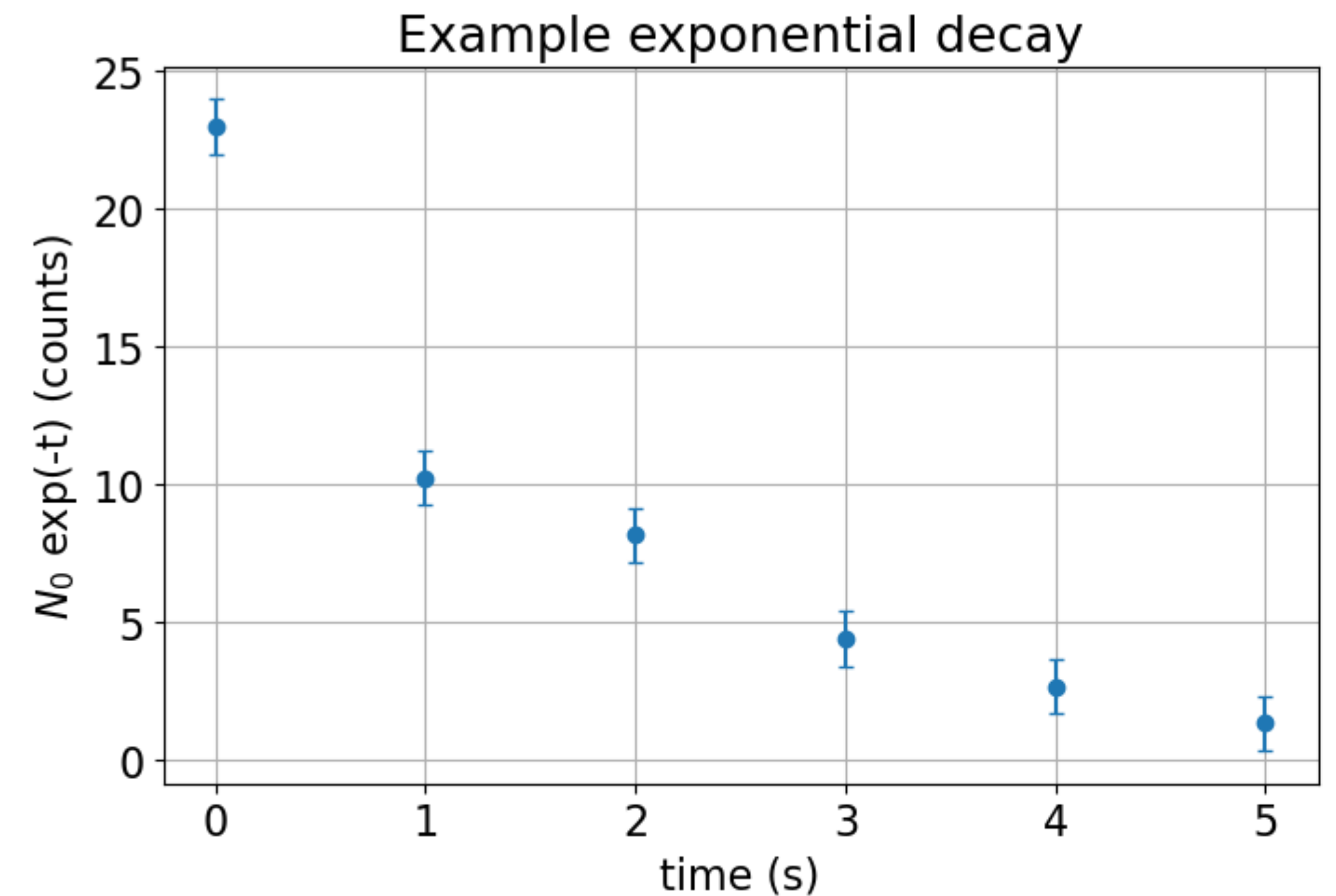
Table of contents

- Introduction
- Measurement Probability Distributions:
- Propagation of Errors and the Variance-Covariance Matrix:
- Curve Fitting:
- (Computer) Matrix Propagaion of Errors:
- Chi-squared Test for Goodness of fit:

Display a menu

Graphs and Presentation of Data

- Please make **graphs** a **reasonable size** in your report!
- Make sure to:
 - include a **title or caption**
 - **label the axes** and include units if appropriate
 - use an **appropriate font size** so that all text is clearly legible
- **Experimental data points** should be presented as points, preferably **with error bars**, and not connected by lines.
- Grid lines may be used.
- LaTeX can be used in titles and labels.
- Save as PDF for inclusion in reports.
- See HowTos.

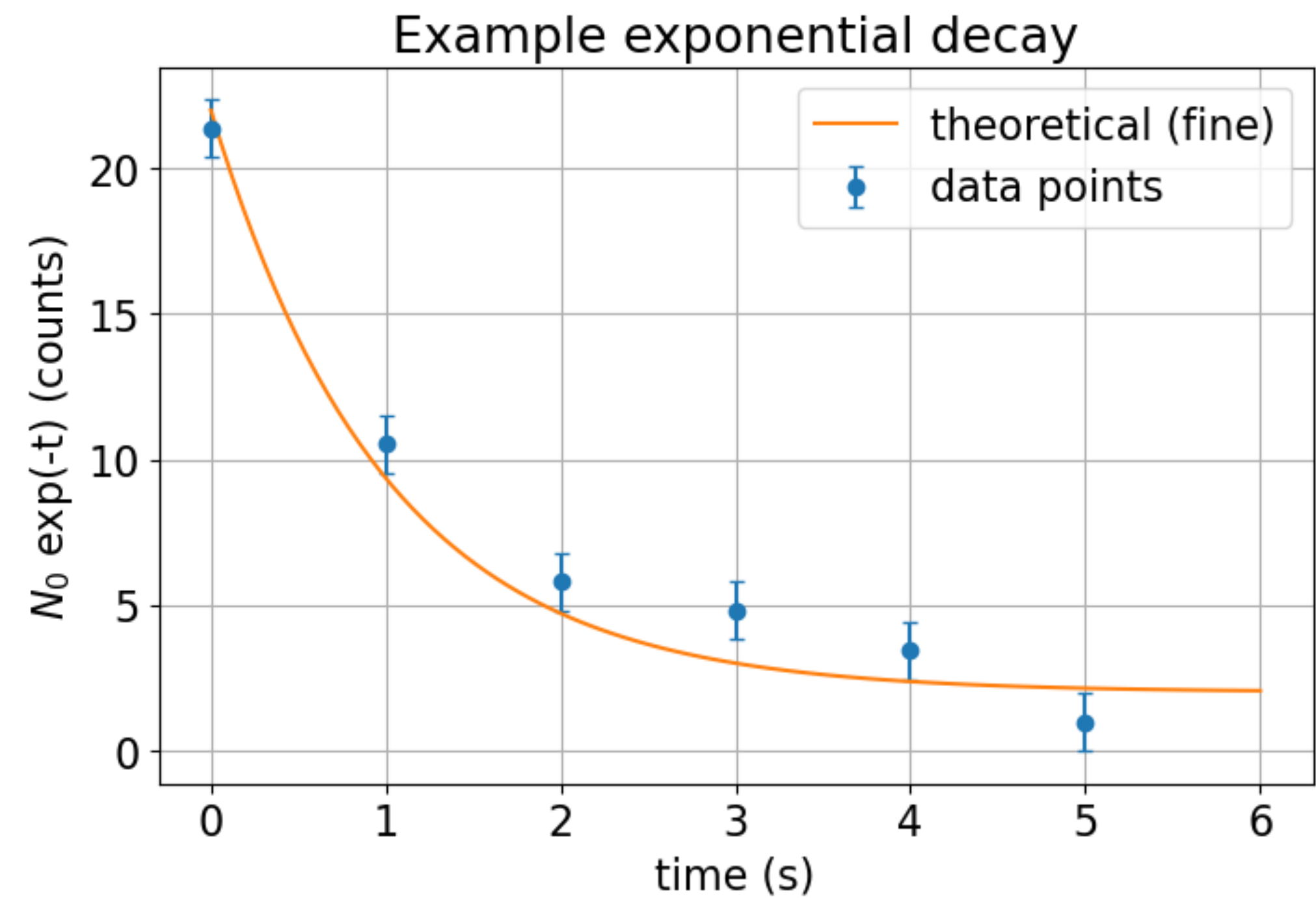
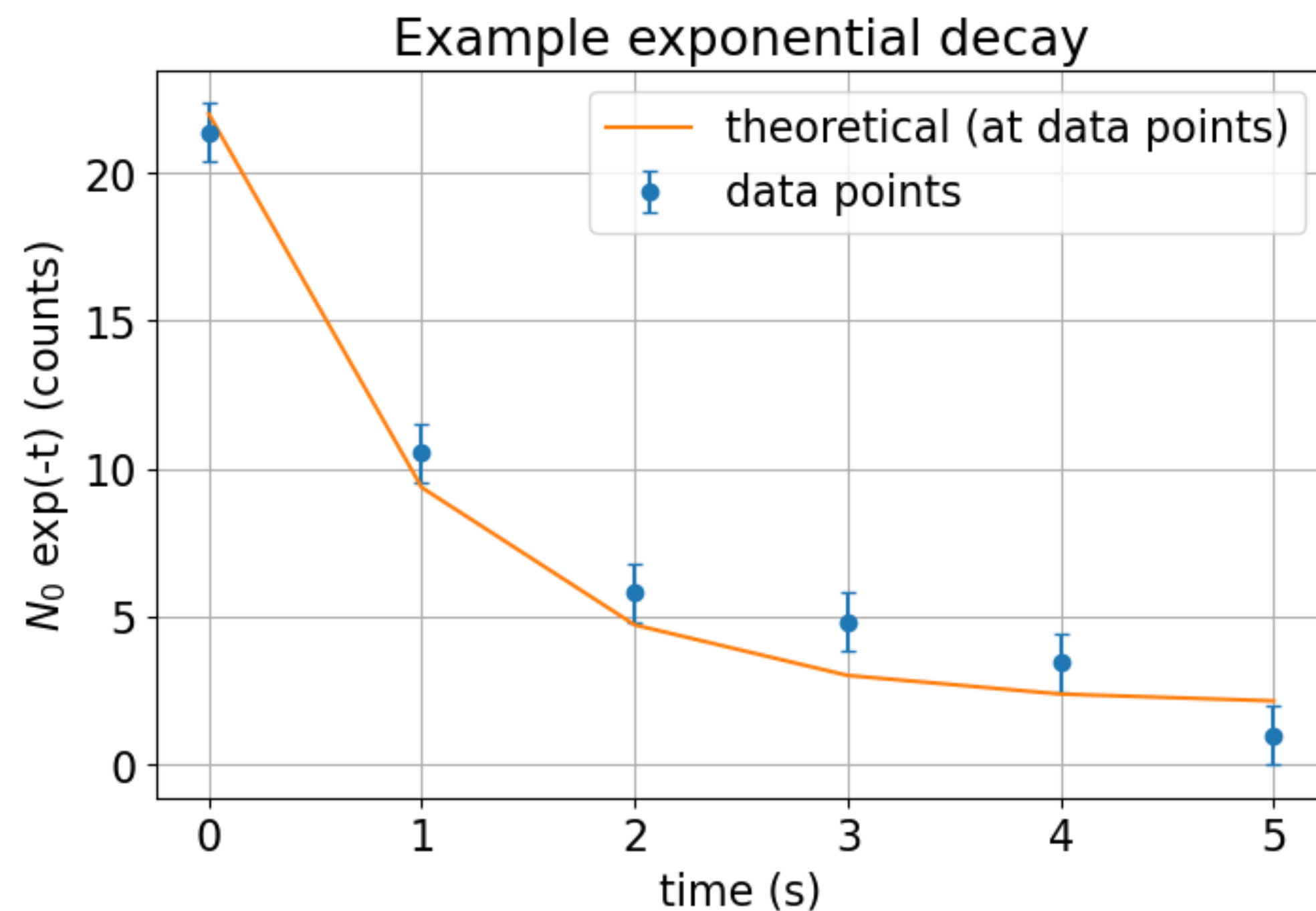


Matplotlib pyplot (plt) - set sizes for notebook:

```
plt.rcParams['figure.figsize'] = (6,4)
plt.rcParams['font.size'] = 12
plt.rcParams['savefig.bbox'] = 'tight'
```

Graphs and Presentation of Data

- Theoretical **curves** (incl. best-fit) should be **overlaid as continuous curves**, possibly with a **finer sampling than the data points**.
- If more than one set of data points/curves use legends to distinguish.



Measurement Errors: Why are they important?

- Other than the 7 defining constants in Physics, every physical quantity has an uncertainty.
 - e.g. NIST: <https://pml.nist.gov/cuu/Constants/Table/allascii.txt>
- Every measurement has an uncertainty:
 - We desire to place a confidence level as to where the true value of the measured value lies.
 - Calculations with uncertain values means we must propagate the uncertainty through the calculations.
- Models and data (even data and data!) never agree exactly.
 - Uncertainties:
 - allow us to quantify the probability that the data and model agree
 - allow use to put uncertainties on model parameters derived from the data.
 - are modelled using probability distributions (All models are wrong, some are useful!)
- The real world is complicated - uncertainty analysis may not be trivial.

Measurement Errors: Why are they important?

Errors in the physical sciences

What is the role of experiments in the physical sciences? In his *Lectures on Physics*, Volume 1, page 1, the Nobel Laureate Richard Feynman states (Feynman 1963):

The principle of science, the definition, almost, is the following: *The test of all knowledge is experiment*. Experiment is the *sole judge* of scientific ‘truth’.

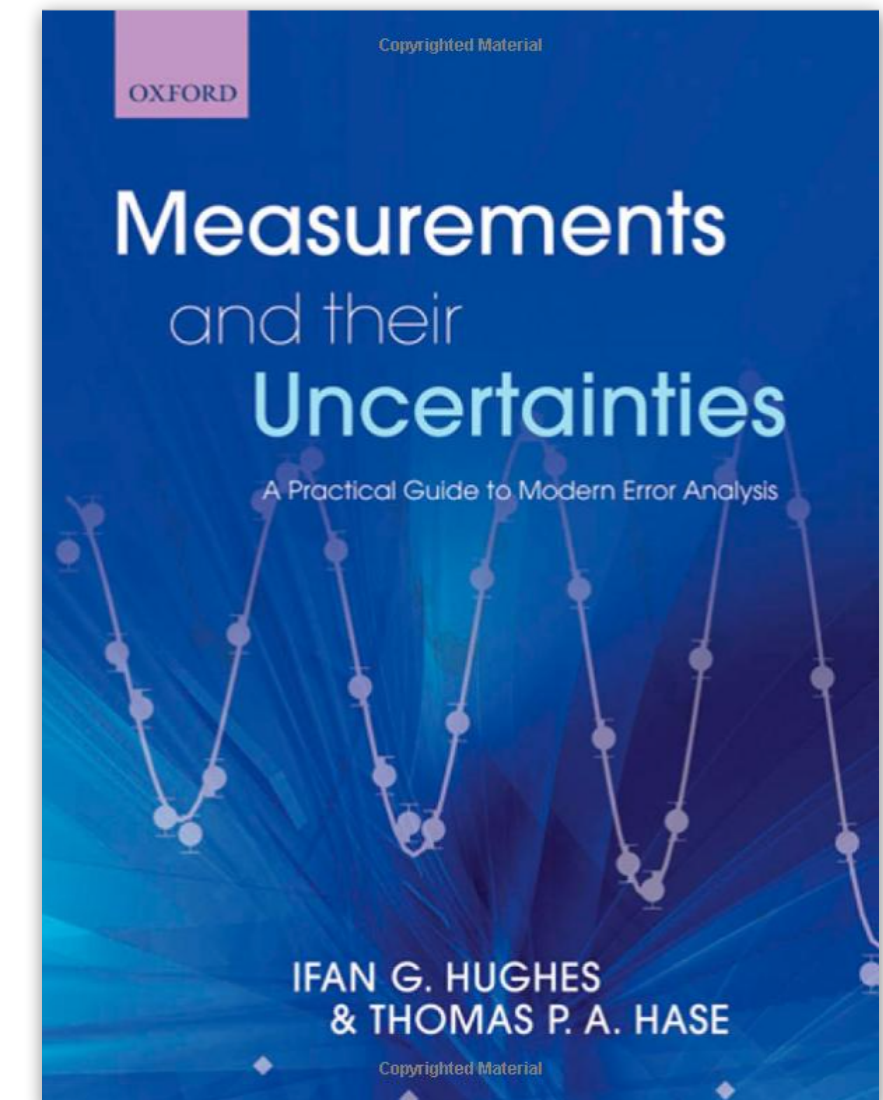
We will emphasise in this book that an experiment is not complete until an analysis of the uncertainty in the final result to be reported has been conducted. Important questions such as:

- do the results agree with theory?
- are the results reproducible?
- has a new phenomenon or effect been observed?

can only be answered after appropriate error analysis.

1

1.1 The importance of error analysis	1
1.2 Uncertainties in measurement	2
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Measurement Errors: Why are they important?

- General Relativity predicts bending of light by gravity.
 - Experiment: measure bending of light by the sun during an eclipse.
 - Newton Prediction: 0.87 arcsec
 - Einstein Prediction: 1.75 arcsec
 - Experimental measurement: (1.98 ± 0.12) arcsec. (0.00055 ± 0.00003 deg.!)
- Strongly support General Relativity and rule out Newtonian prediction.!

PHILOSOPHICAL TRANSACTIONS OF THE ROYAL SOCIETY A MATHEMATICAL, PHYSICAL & ENGINEERING SCIENCES

A Determination of the Deflection of Light by the Sun's Gravitational Field, from Observations Made at the Total Eclipse of May 29, 1919

F. W. Dyson, A. S. Eddington and C. Davidson

Phil. Trans. R. Soc. Lond. A 1920 **220**, doi: 10.1098/rsta.1920.0009, published 1 January 1920

<https://royalsocietypublishing.org/doi/10.1098/rsta.1920.0009>

Measurement Errors

- **Statistical (or Random) Errors:**

- repeat experiment multiple times - values fluctuate.
- we assume they fluctuate about some mean value
- we assume random errors average to 0 over many repeated measurements.
- ultimate limit: resolution of instrument

- **Systematic and Non-Statistical Errors:**

- additional uncertainties due to incomplete/imperfect/incorrect knowledge of experiment/calibration etc.
- Not easy to detect and correct.
- Generally lumped together into the term "Systematics"

- **In general when we quote errors on an experiment they are the Statistical Errors.**

- Non-Statistical/Systematic Errors may be quoted in addition: , e.g. $x = 1.0 \pm 0.1_{stat} \pm 0.2_{sys}$

Measurement Uncertainty

- **Measurement Errors:**

- **Type A:** derive from statistical analysis of repeated measurements
- **Type B:** non-statistical, using other information, e.g. from instrument specification.

- **Instrumentation Precision:**

- **Analogue:** The statistical error associated with analogue instrumentation is often due to how well one can read the scale, and it is often up to the experimenter to estimate.
- **Digital Meter:** Guide: the precision of a digital meter is limited to the last digit.
 - e.g. repeated measurements of a voltage with a digital multimeter gives 8.41 V. We would thus quote the voltage as 8.41 ± 0.01 V.
- **ADC:** resolution limited by number of bits, i.e. range divided by 2^N where N is the number of bits. ADCs round down!
- **Other:** instrumentation documentation
- **Non-instrumental:**
 - noise or other errors may be greater than instrument precision
 - e.g. measure repeated time intervals with a precision stop watch.

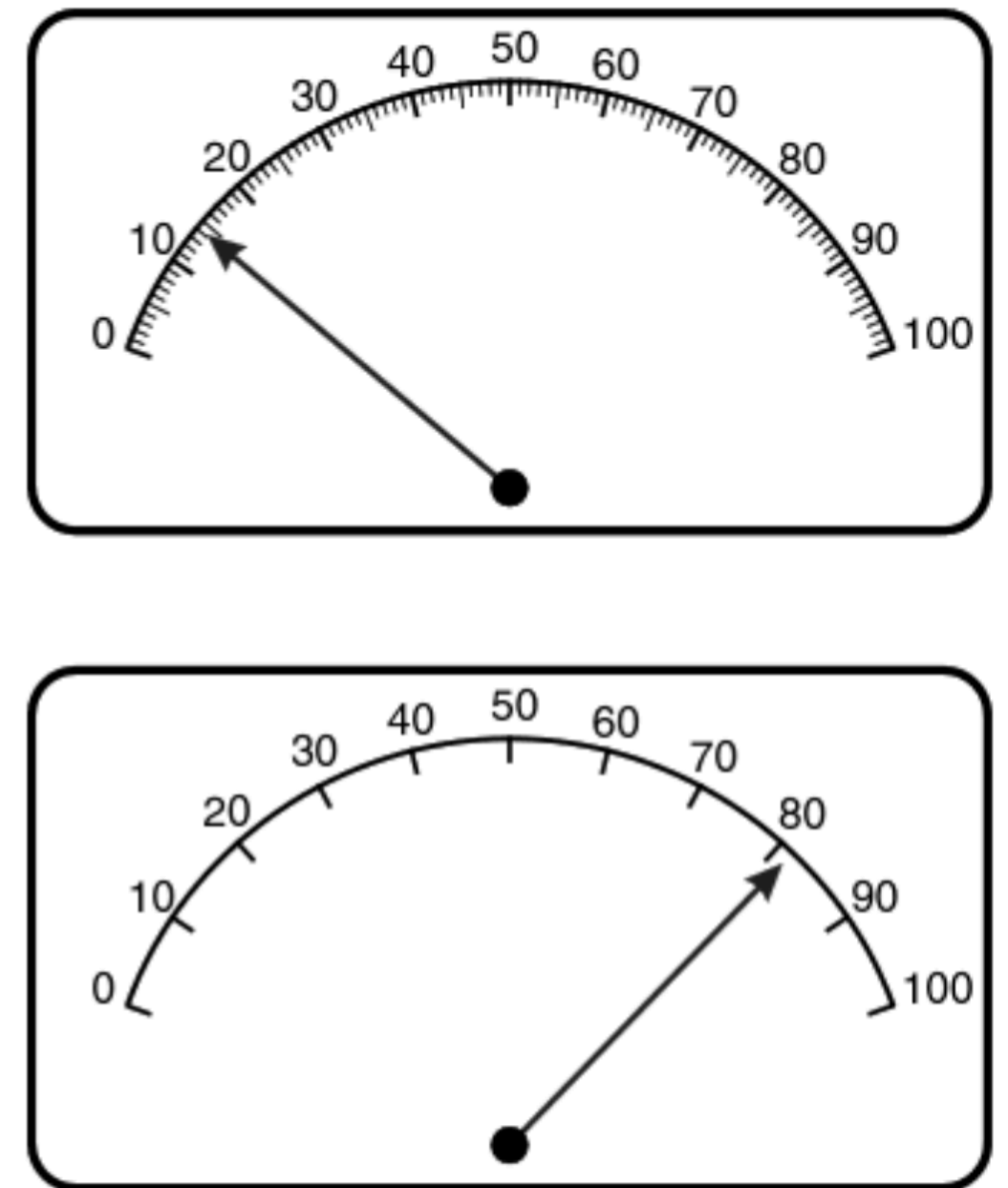


Fig. 1.4 The upper part of the figure displays a situation where estimating the uncertainty to be half a division is appropriate; in contrast in the lower part of the figure the uncertainty in the measurement is substantially smaller than half a division.

(from Hughes & Hase book)

Mean and Standard Deviation

For data, the mean, μ , and (sample) standard deviation, σ , can be calculated from the data as:

$$\mu = \bar{x} = \frac{1}{N} \sum_{i=1}^N x_i$$

and

$$\sigma^2 = \frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2$$

(if deriving the mean and the variance/standard deviation from the data)

Probability Distributions

- In Physics experiments/observations the following probability distributions are commonly met:
 - **Binomial** - when two possible outcomes- **not covered in these tutorials**.
 - **Gaussian** (also called **Normal**) - common distribution in nature.
 - Note: the Gaussian distribution is the limiting case of the Binomial and Poisson distributions - generally we want to be in the Gaussian regime!
 - **Poisson** - counting rare events
 - **Student's t** - **not covered in these tutorials**.
- In comparing Physics measurements to a model we also meet: χ^2 .
- Note: `scipy.stats` has every statistical distribution you could want! [link](#)

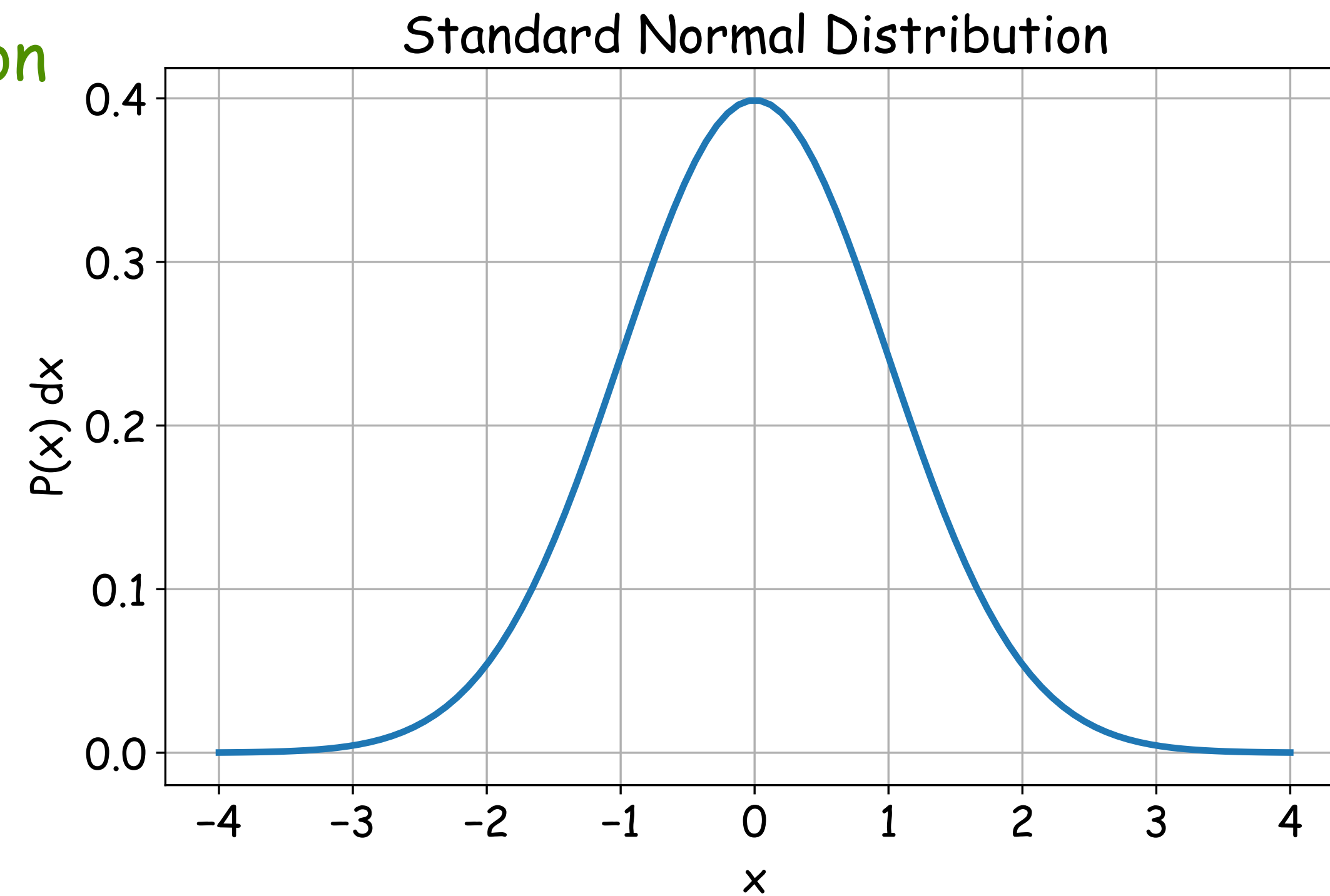
The Gaussian (Normal) Distribution

- Statistical Errors tend to follow a Gaussian Distribution

$$P(x) dx = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$$

where:

- $P(x)$ is a probability density function.
- $P(x) dx$ is the probability of obtaining a value between x and $x + dx$.
- μ is the mean (centre) of the distribution
- σ is a measure of the width of the distribution
- normalised: area = 1.

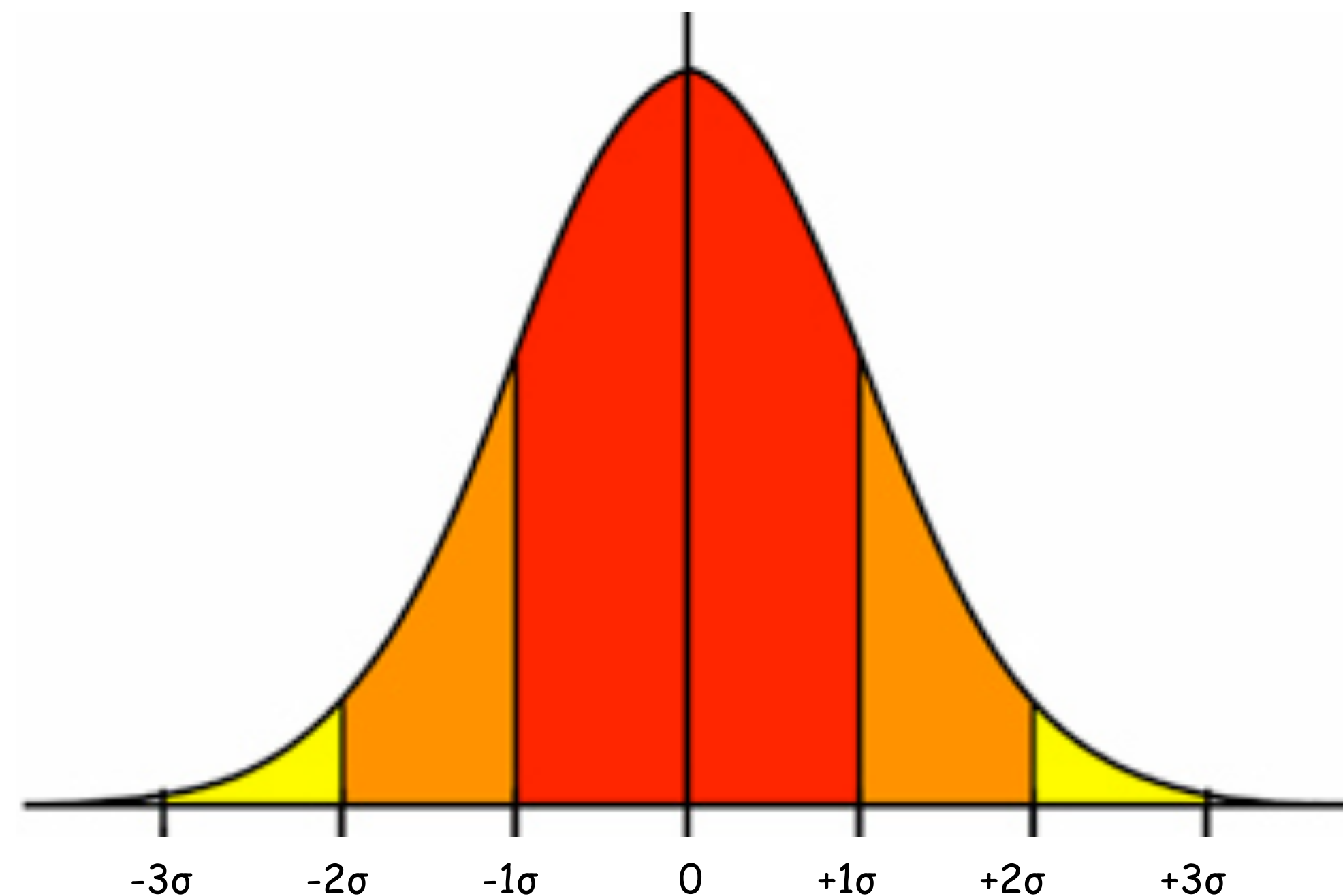


Standard Normal: $\mu=0, \sigma=1$

The Normal distribution is the limiting case of the Poisson distribution when $\mu \geq 35$ (note: rough $\mu \gtrsim 10$).

The Gaussian (Normal) Distribution

- If we know the mean and standard deviation for a set of measurements (or technique) **normally distributed**, what is the probability that a measurement will fall in a given range?



|← 68.3% →|

~1 in 3 outside of range

|← 95.4% →|

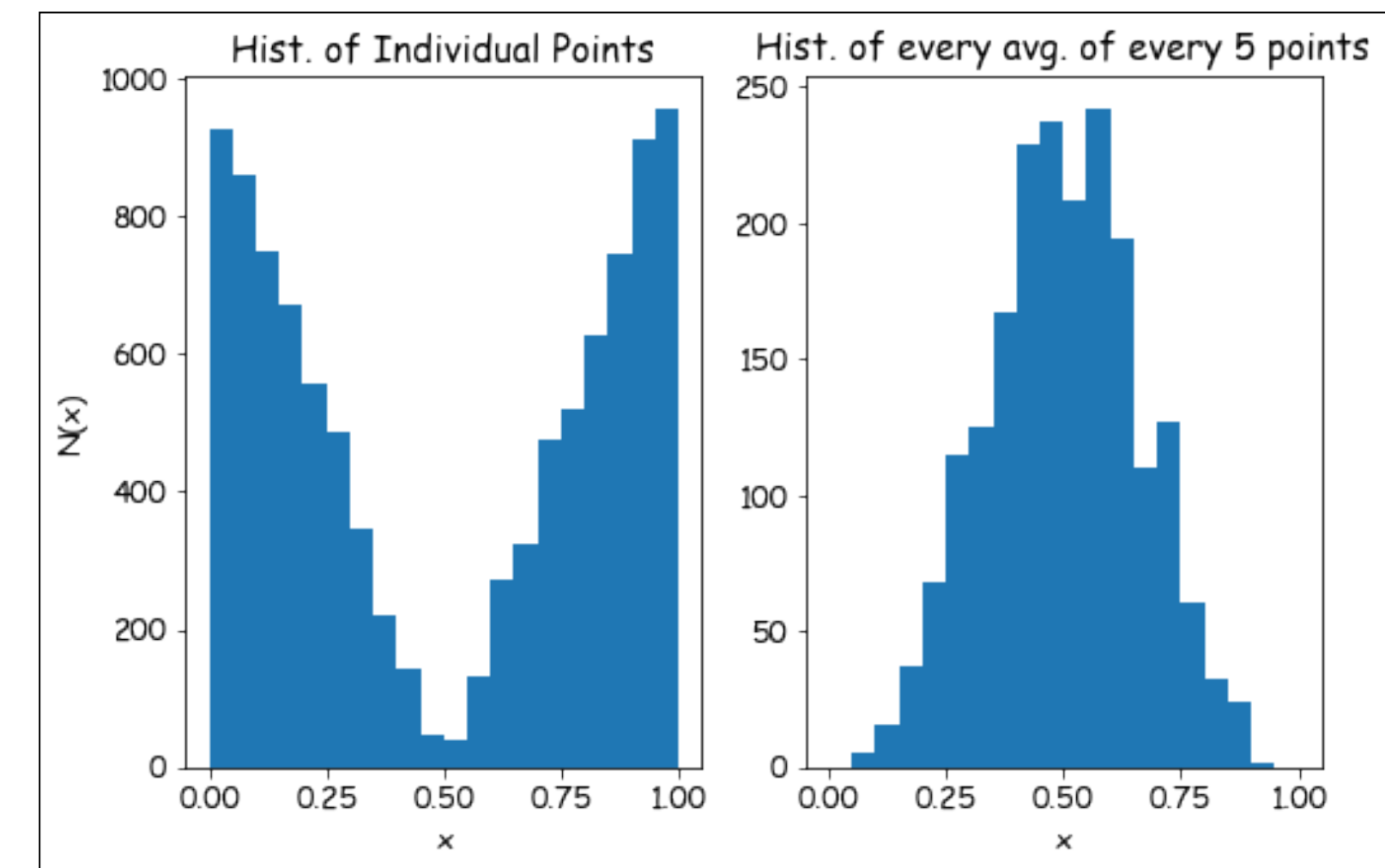
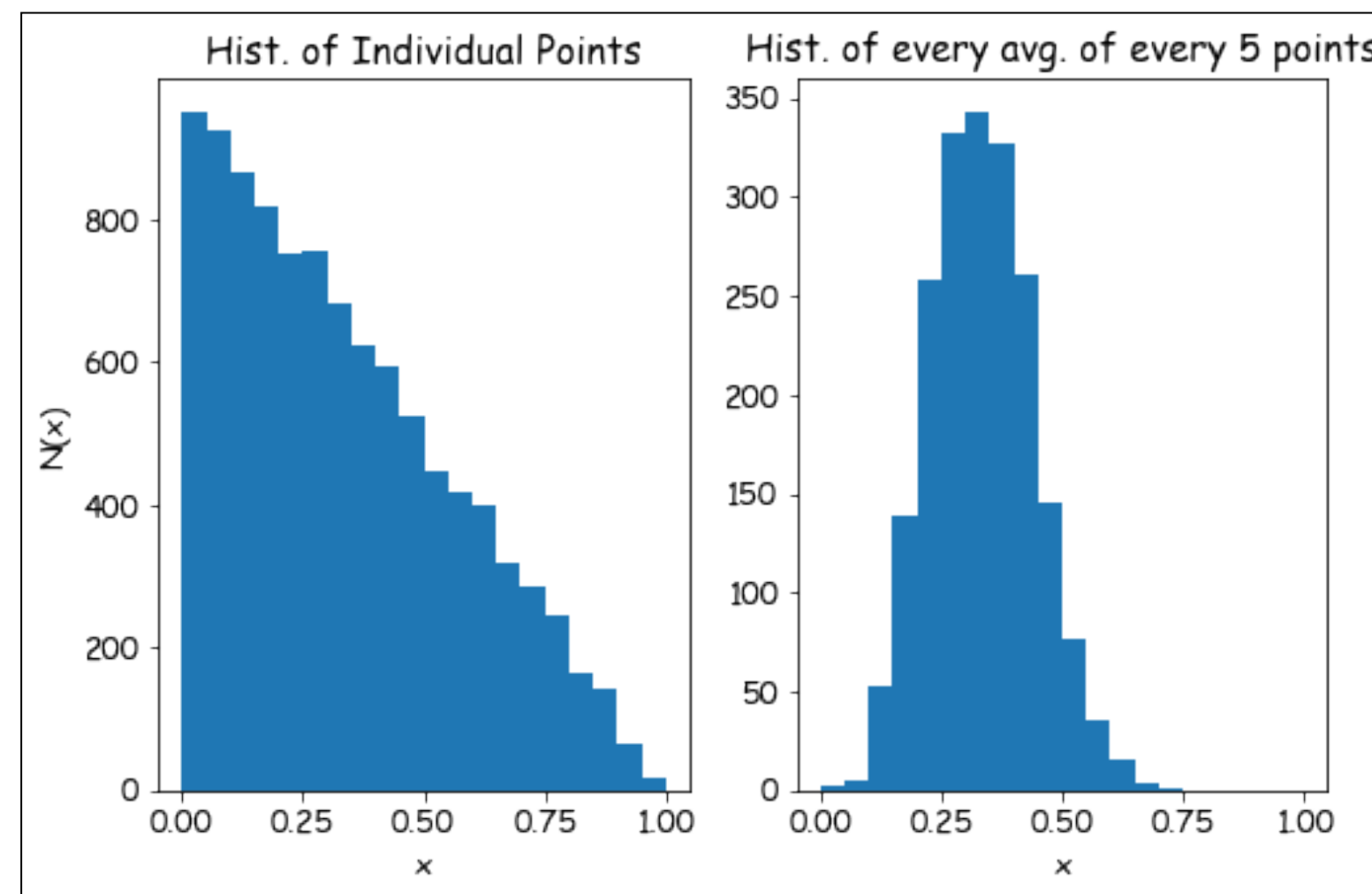
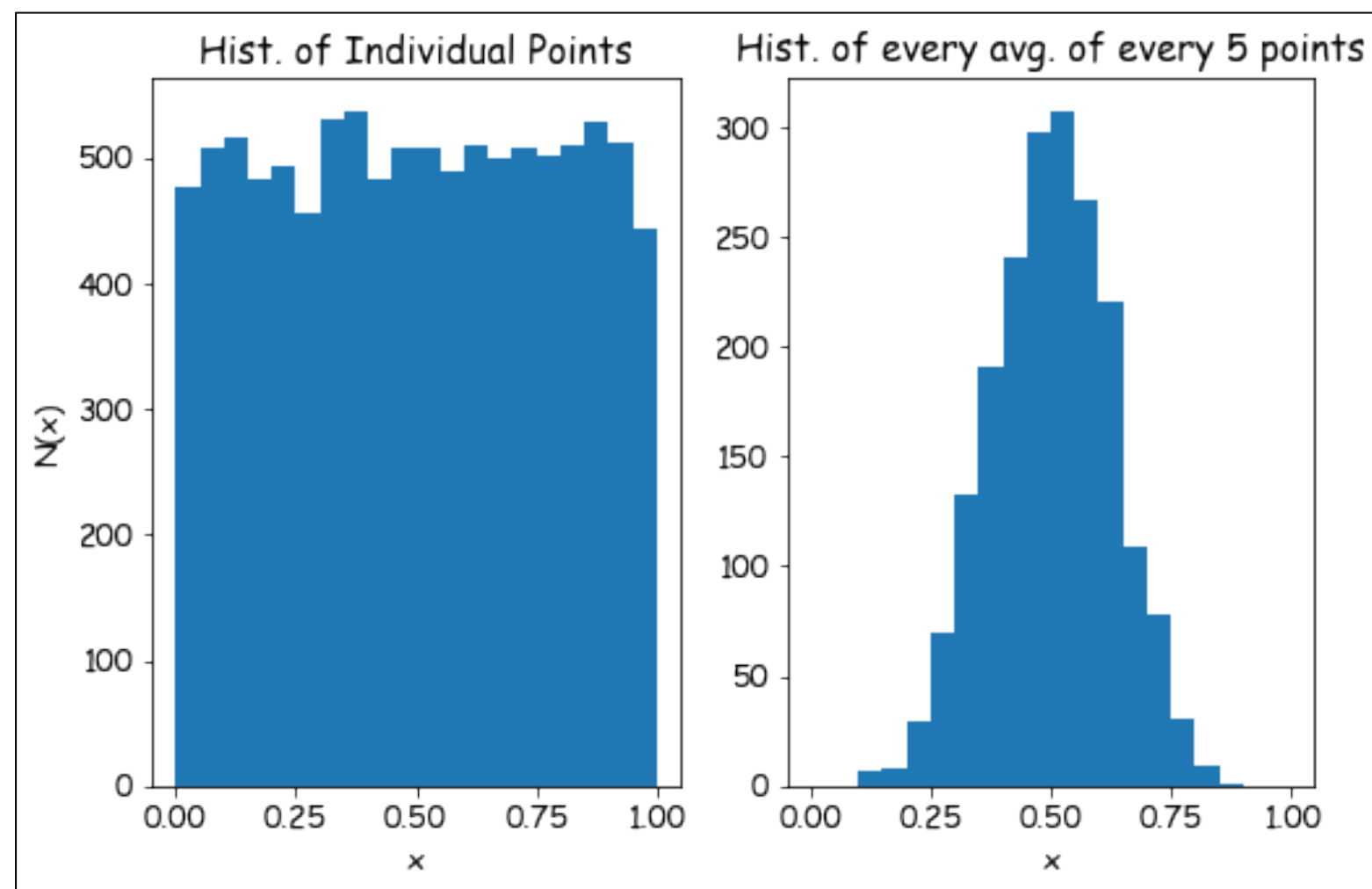
~1 in 20 outside of range

|← 99.7% →|

~1 in 400 outside of range

Central Limit Theorem

- Why is the normal distribution so prevalent?
 - Ans: The Central Limit Theorem
 - Irrespective of the parent distribution of some variable, the distribution of the sum or mean of that variable tends towards a normal distribution with the same mean, as the number of samples becomes large (not many needed for many 'reasonable functions'!)



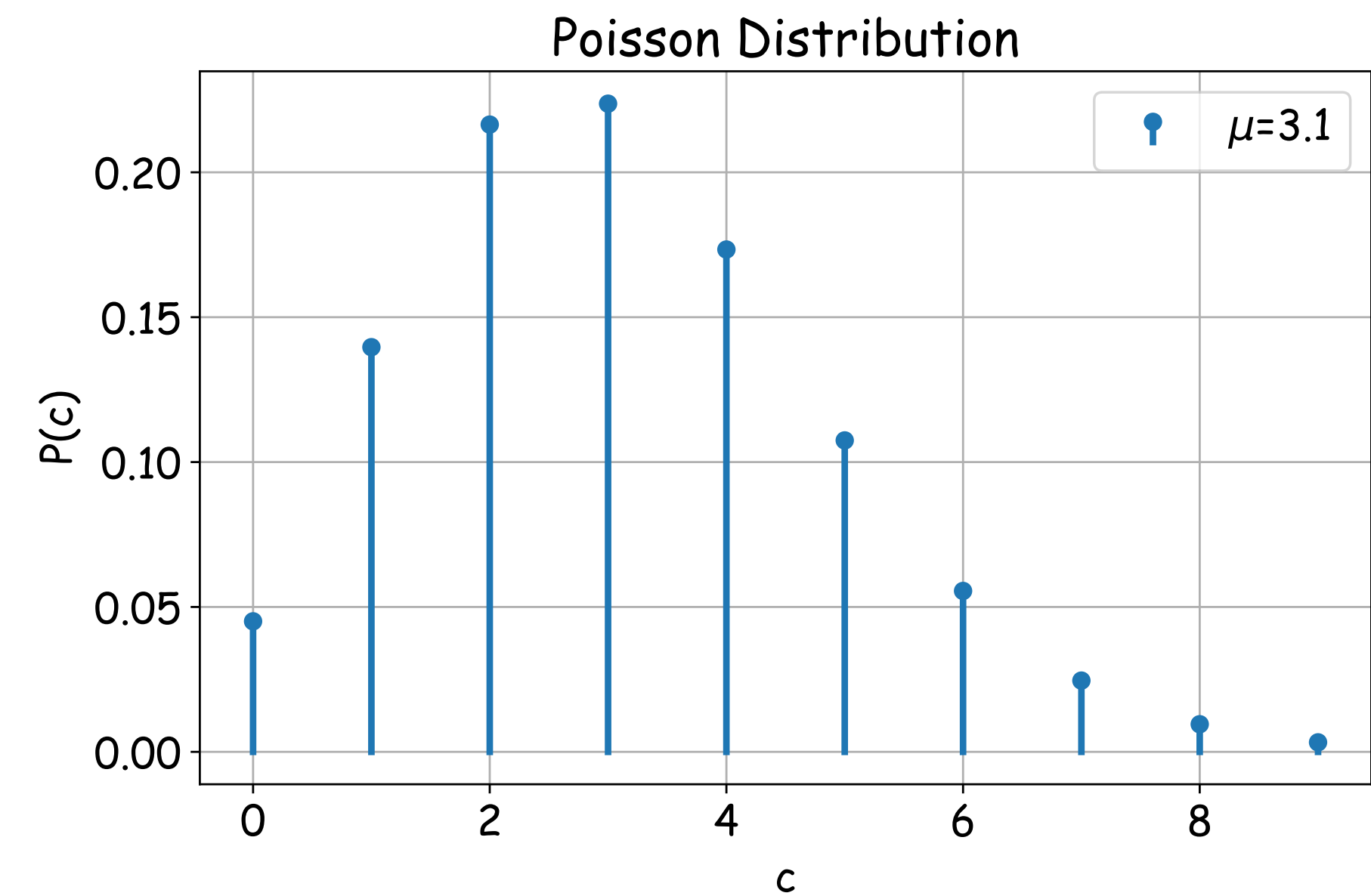
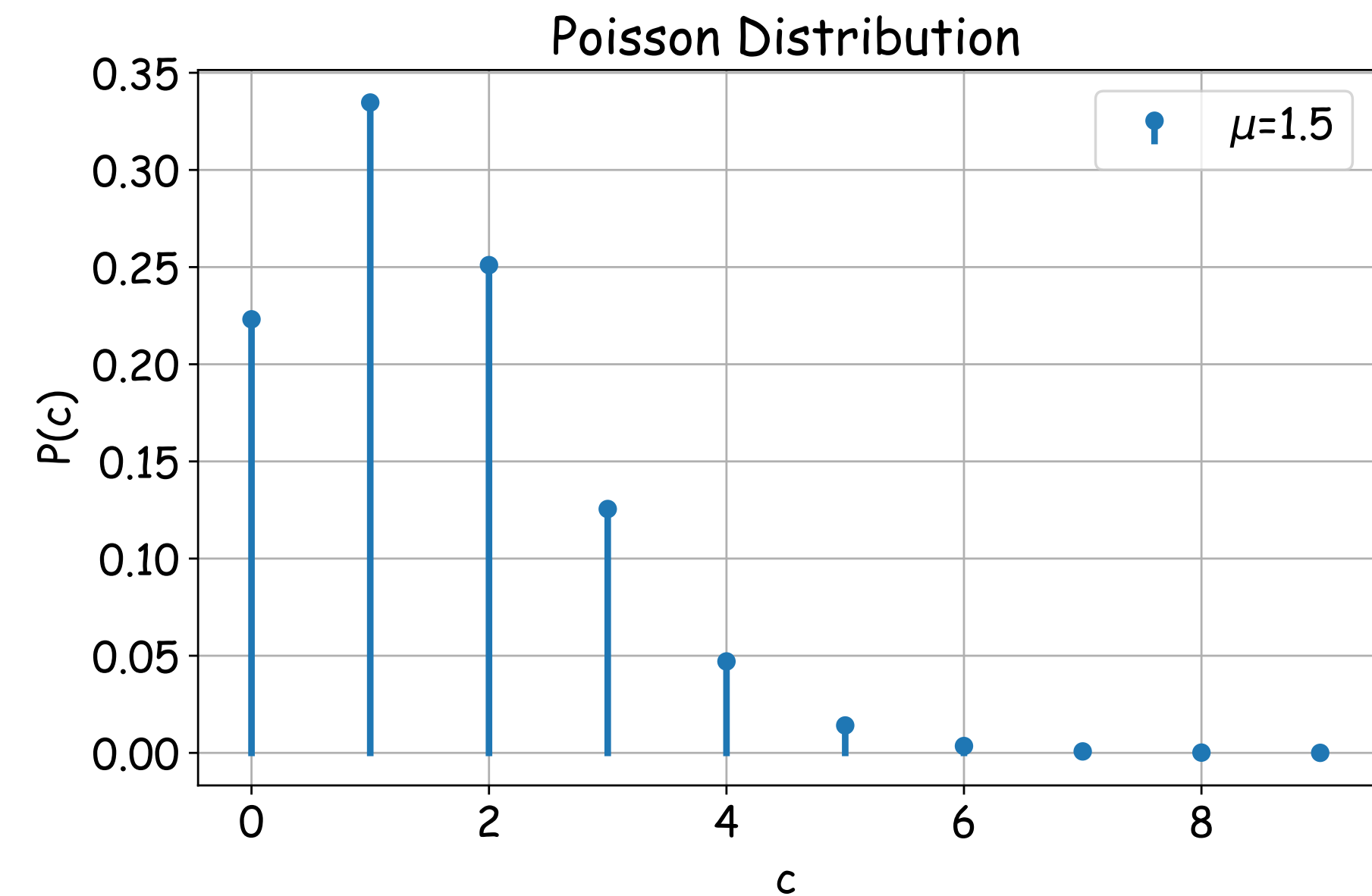
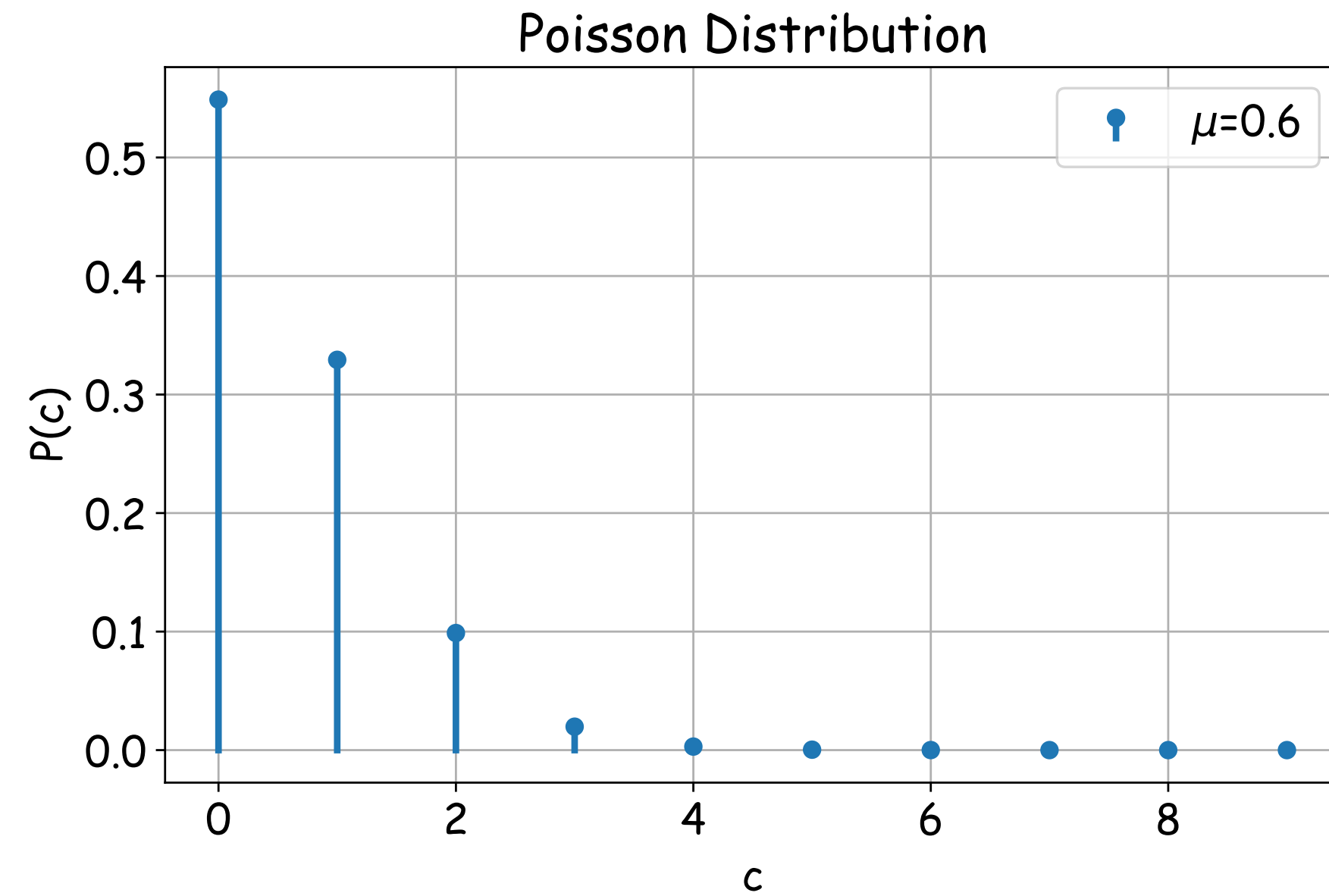
Poisson Distribution

- Poisson (discrete distribution)
 - When one counts the number of random events in an interval (time, area, volume, etc) and repeats the experiment under identical conditions then one does not always get the same result.
 - The distribution of counted events follows a Poisson Distribution:

$$P(n) = \frac{\mu^n e^{-\mu}}{n!}$$

- where,
 - $P(n)$ is the probability of obtaining n events in a given interval
 - μ is the mean of the distribution.
 - the standard deviation has value $\sigma = \sqrt{\mu}$
- The Poisson distribution is applicable to many experiments involving counting rare events such as radioactive decay, photons etc. (only μ measureable)

Poisson Distribution



Asymmetric!

$$\mu = 0.6$$

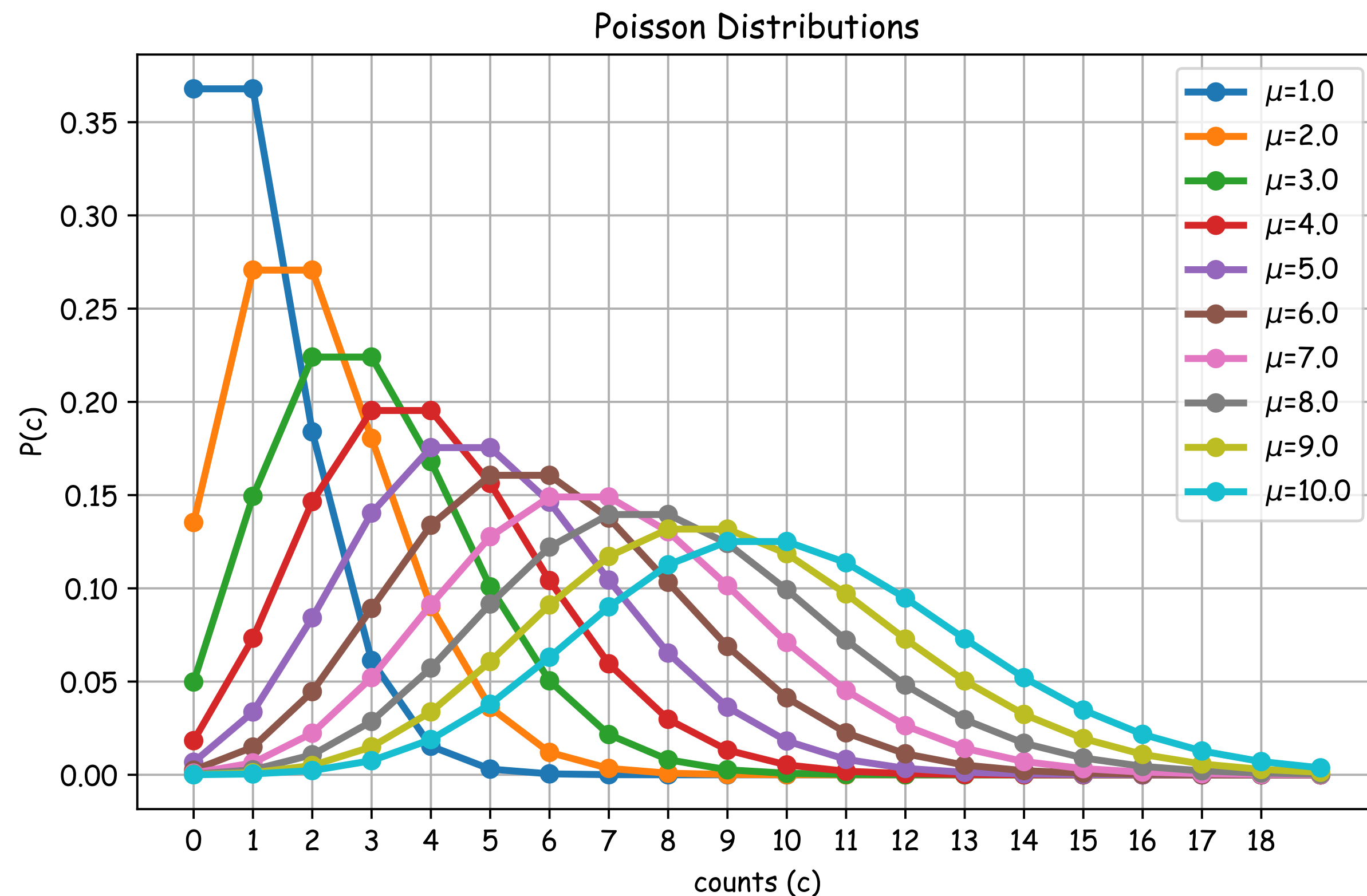
$$\sigma = \sqrt{0.6} = 0.775$$

It does not make sense to quote: 0.600 ± 0.775

We need symmetric error bars!

Poisson Distribution

- For **small means the distribution is asymmetric**.
- For **means $\mu \gtrsim 10$ the distribution is nearly symmetric** and is approximately described by a Gaussian distribution of mean μ and standard deviation $\sqrt{\mu}$.



- For dealing with errors (especially propagation) in counting experiments **we generally want enough counts for the Poisson distribution to be in Gaussian regime**. (then Gaussian with $\sigma = \sqrt{\mu}$!)

Propagation of Errors

- Assume we have made some measurements (e.g. length and width) and **we want to combine the measurements using some formulae to calculate a some property** (for example, area).
- How do we estimate the uncertainty on the final result given we know the uncertainties on the initial measurements? (i.e. what is $A \pm dA$?)
- Use **Propagation of Errors Formula** (without proof) **for Gaussian errors**:
 - Say x is calculated using values with uncertainties $u, v, \dots$ (i.e. $x = f(u, v, \dots)$)
 - Then, for uncorrelated measurement fluctuations:

$$\sigma_x^2 = \sigma_u^2 \left(\frac{\partial x}{\partial u} \right)^2 + \sigma_v^2 \left(\frac{\partial x}{\partial v} \right)^2 + \dots$$

- If the **measurement fluctuations are correlated** then we have to **include an additional term called the covariance**. (no time to cover covariance in these tutorials!)
- (covariance critical for confidence region on best-fit parameters)

Propagation of Errors: some common formulae

$$x = av \pm bv$$

$$\sigma_x^2 = a^2\sigma_u^2 + b^2\sigma_v^2 \pm \cancel{2ab\sigma_{uv}^2}$$

$$x = \pm auv$$

$$\frac{\sigma_x^2}{x^2} = \frac{\sigma_u^2}{u^2} + \frac{\sigma_v^2}{v^2} + \cancel{\frac{2\sigma_{uv}}{uv}}$$

$$x = \pm \frac{au}{v}$$

$$\frac{\sigma_x^2}{x^2} = \frac{\sigma_u^2}{u^2} + \frac{\sigma_v^2}{v^2} - \cancel{\frac{2\sigma_{uv}}{uv}}$$

$$x = au^{\pm b}$$

$$\frac{\sigma_x}{x} = \pm b \frac{\sigma_u}{u}$$

$$x = ae^{\pm bu}$$

$$\frac{\sigma_x}{x} = \pm b\sigma_u$$

$$x = a^{\pm bu}$$

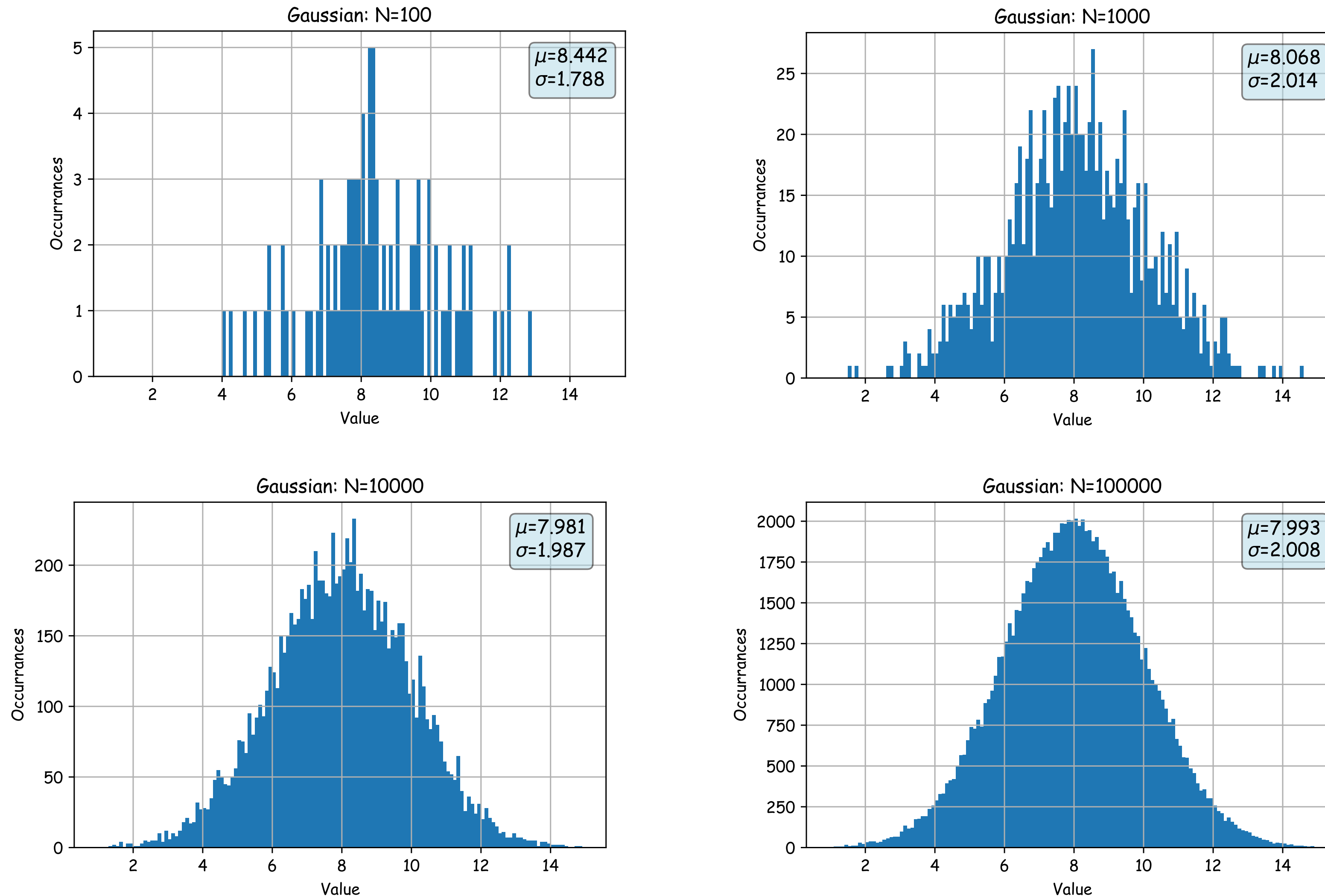
$$\frac{\sigma_x}{x} = \pm (b \ln a) \sigma_u$$

$$x = a \ln(\pm bu)$$

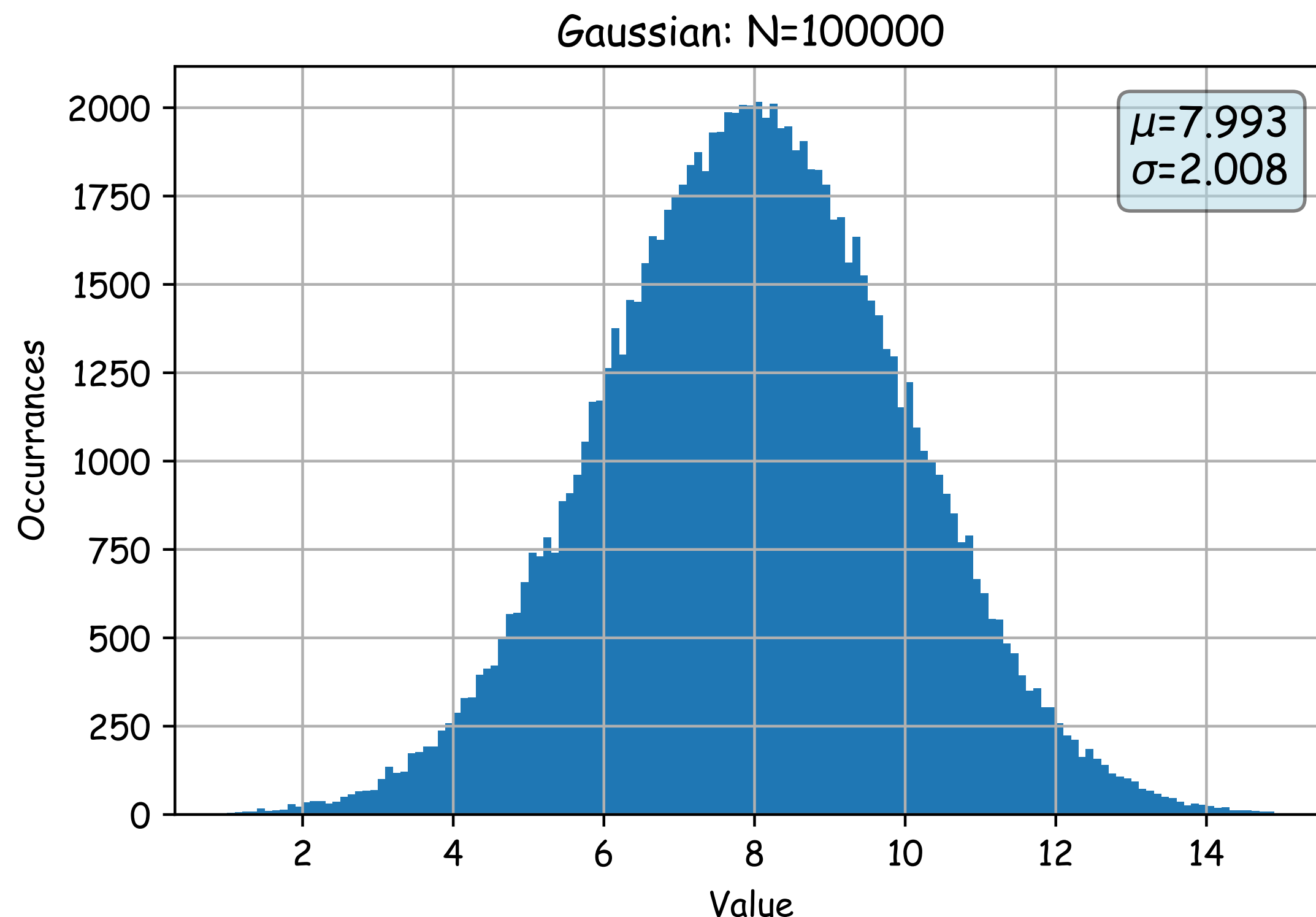
$$\sigma_x = a \frac{\sigma_u}{u}$$

The Mean and Its Uncertainty

- The more measurements of a quantity we make the more precisely we can characterise the distribution



The Mean and Its Uncertainty



- What does this distribution tell us?
 - The probability of getting a given value in a single measurement.
- How well do we know the mean / what is the uncertainty on the mean?
 - We know the mean much more accurately than to within $\pm 1\sigma$ of the parent distribution!
- Using Propagation of Errors it is possible to show that the uncertainty on the mean is:

$$\sigma_{\mu} = \frac{\sigma}{\sqrt{N}}$$

If N data points are averaged to get the mean, the error on the mean is not the standard deviation σ of the distribution but $\sigma/\sqrt{N}$ ("the standard error on the mean")

Quoting Errors

- Please read Sections 2.8 & 2.9 of Hughes & Hase book!
- General guidance:
 - The best estimate of a parameter is the mean.
 - The error is the standard error on the mean.
 - Perform calculations using all significant figures.
 - Errors are generally only quoted to 1 significant place (unless: lots and lots of data has been used to determine the error or the first significant figure is a "1" - then use two)
 - Match the number of places in the mean (or parameter being quoted after error propagation) to the error.
 - e.g. $(96.8645 \pm 0.2701) \times 10^3 \Omega$ should be quoted as $(96.9 \pm 0.3) \times 10^3 \Omega$
 - e.g. $(96.8645 \pm 0.1435) \times 10^3 \Omega$ should be quoted as $(96.86 \pm 0.14) \times 10^3 \Omega$

Curve Fitting: least-squares and χ^2 fitting

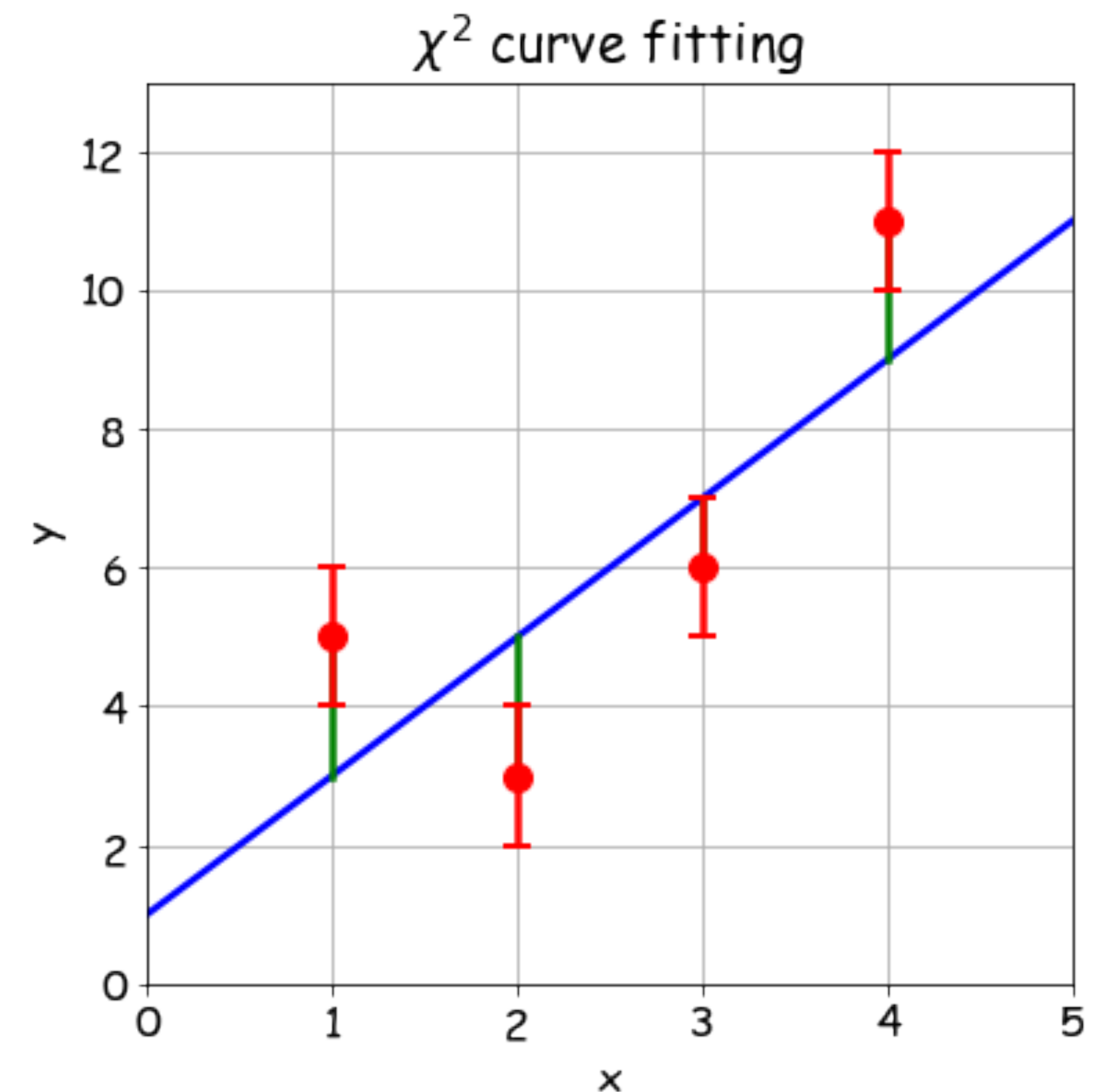
- The **Method of Least Squares** involves adjusting the parameters of a function so that the **sum of deviation of each data point squared is minimised**,
- e.g. fitting a straight line to data we must find m and c which minimise:

$$S = \sum_i [y_i - f(x_i)]^2 = \sum_i [y_i - (mx_i + c)]^2$$

- We can also **include the errors on the data points to 'weight' the points by their errors (Chi-squared (χ^2) minimisation)**:

$$S = \sum_i \left[\frac{y_i - f(x_i)}{\sigma_i} \right]^2 = \sum_i \left[\frac{y_i - (mx_i + c)}{\sigma_i} \right]^2$$

- Software packages such as `scipy.optimize.curve_fit()` perform least squared and χ^2 minimisations.



Using `curve_fit()`

```
from scipy.optimize import curve_fit
```

```
popt, pcov = curve_fit(function,          # Function to fit, required
                        xdata,            # x data points, required
                        ydata,            # y data points, required
                        sigma=yerr,        # Errors on y points, optional
                        absolute_sigma=True, # Errors absolute?, optional
                        p0=initial_pars)    # Initial parameters for search, optional
```

```
popt # best-fit parameters
```

```
pcov # covariance matrix, diagonal are error on each parameter squared
```

- See: https://github.com/UCD-Physics/Python-HowTos/blob/main/Curve_fit.ipynb

χ^2 and Goodness of Fit

- The χ^2 test is a test for Goodness of Fit.
- It can be used to compare data with theory (e.g. a theoretical curve is fit to experimental data and we make a statement about the probability that the theory and data agrees).
- It can also be used to compare two different data sets to see if they agree.

• Definition:

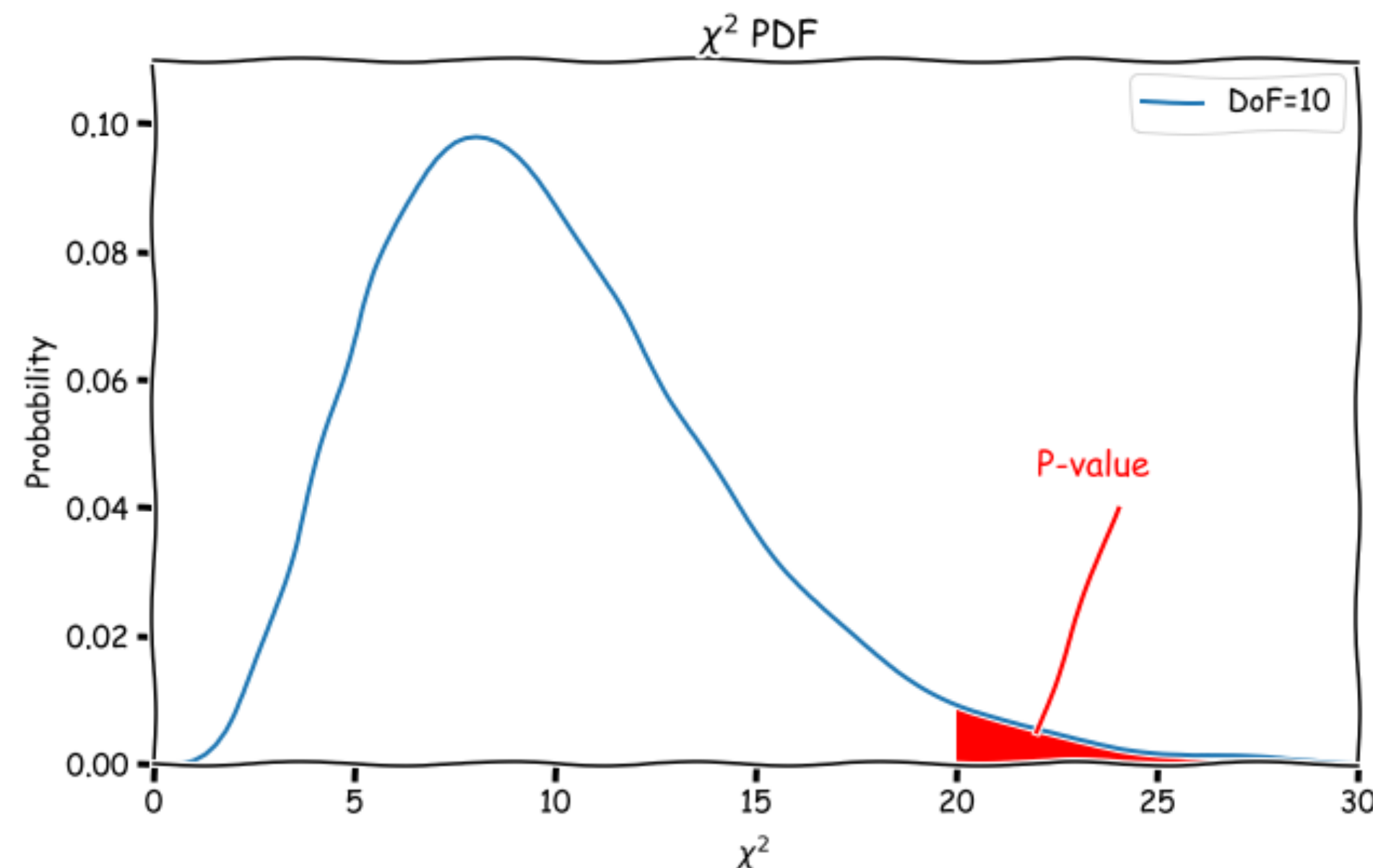
$$\chi^2 \equiv \sum_{i=1}^N \left(\frac{\text{measured}_i - \text{expected}_i}{\text{error on measured}_i} \right)^2$$

[* note different errors for counting experiments]

- If we have reasonably good agreement between data and theory then we would expect that we would get a contribution of ≈ 1 from each data point to sum.
- This is approximately true, in fact we get $\chi^2 \approx \nu = n - n_c$ where ν is called the "Number of Degrees of Freedom" and is equal to the number of data points, n , minus the number of constraints (free parameters in fit function), n_c , derived from the data.
- χ^2/ν is called the "Reduced Chi Squared" and for a good fit is ≈ 1 .

χ^2 and Goodness of Fit

- χ^2 is a **distribution** of $\sum (\text{normal deviates})^2$ with a unique curve for each number of degrees of freedom.
- After finding **the optimum parameters by minimising χ^2** we can **check to see if the χ^2 value is reasonable for the degrees of freedom in question:**
 - the reduced χ^2 gives a good indication (but interpretation depends strongly on the DOF)
 - **better: the P-value:**
 - **the probability of obtaining a value of $\chi^2 \geq \chi^2_{\text{observed}}$ by chance** (for normally distributed errors if the model and data agree) is given by **integrating the χ^2 distribution from the observed value to infinity:**



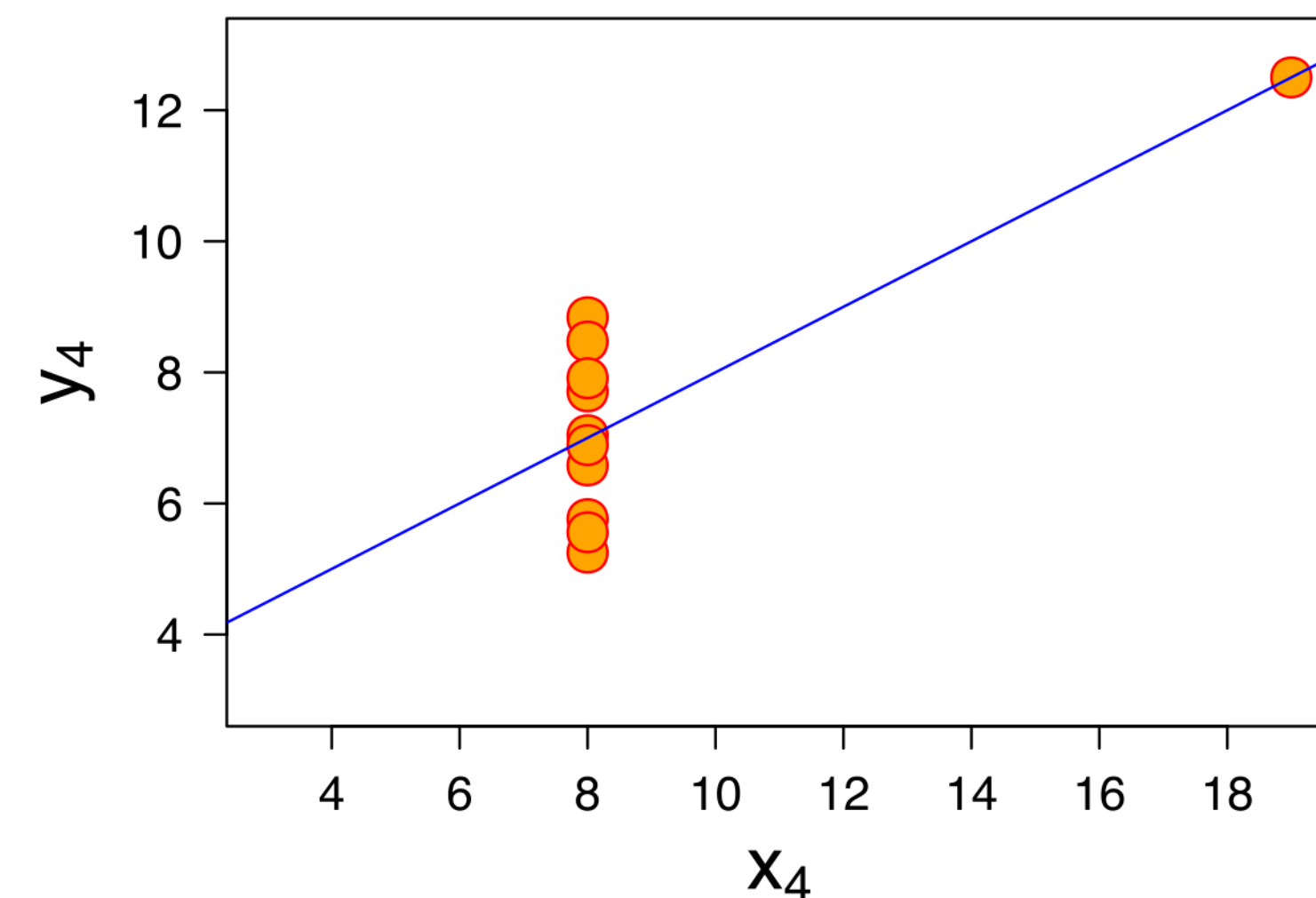
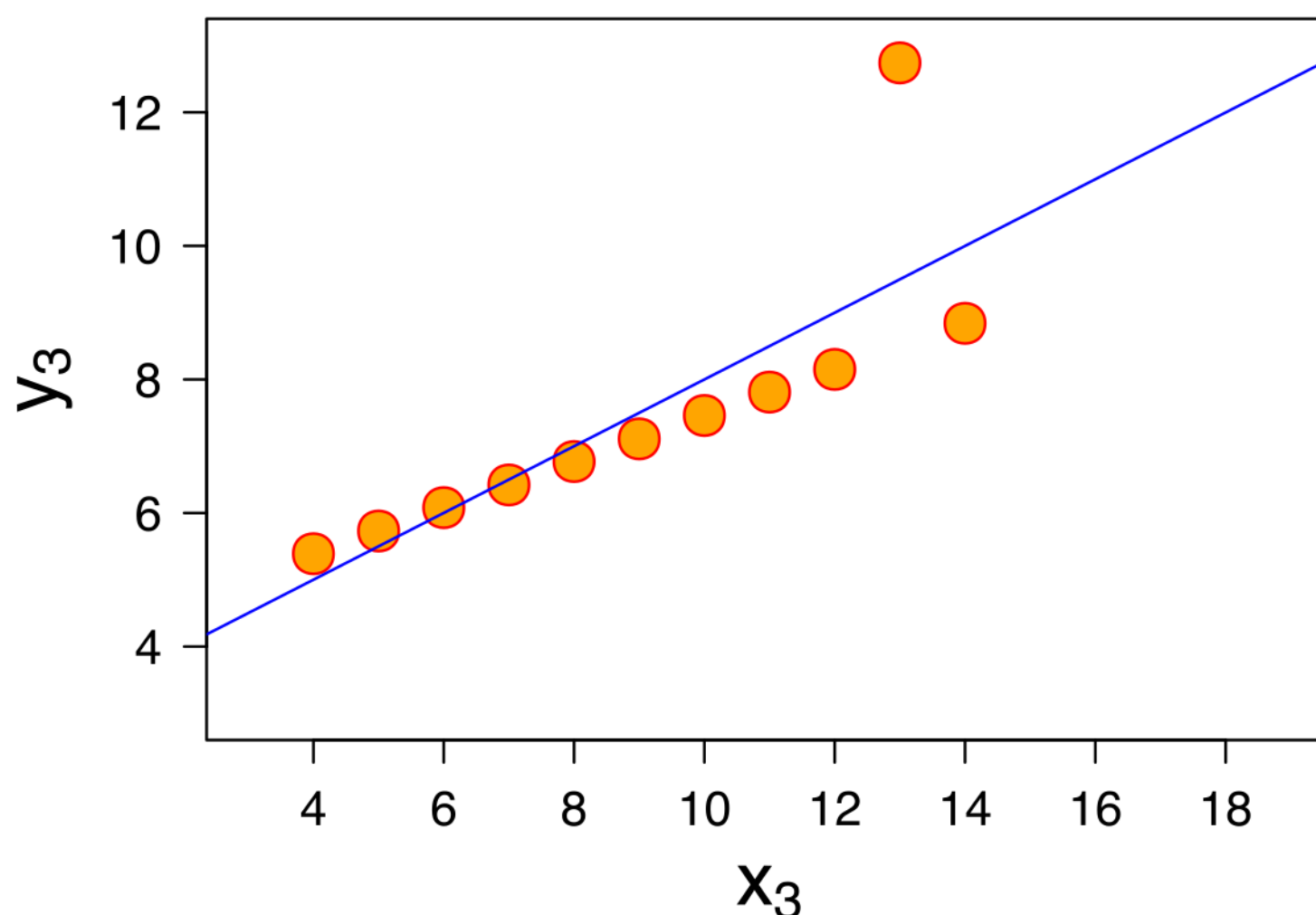
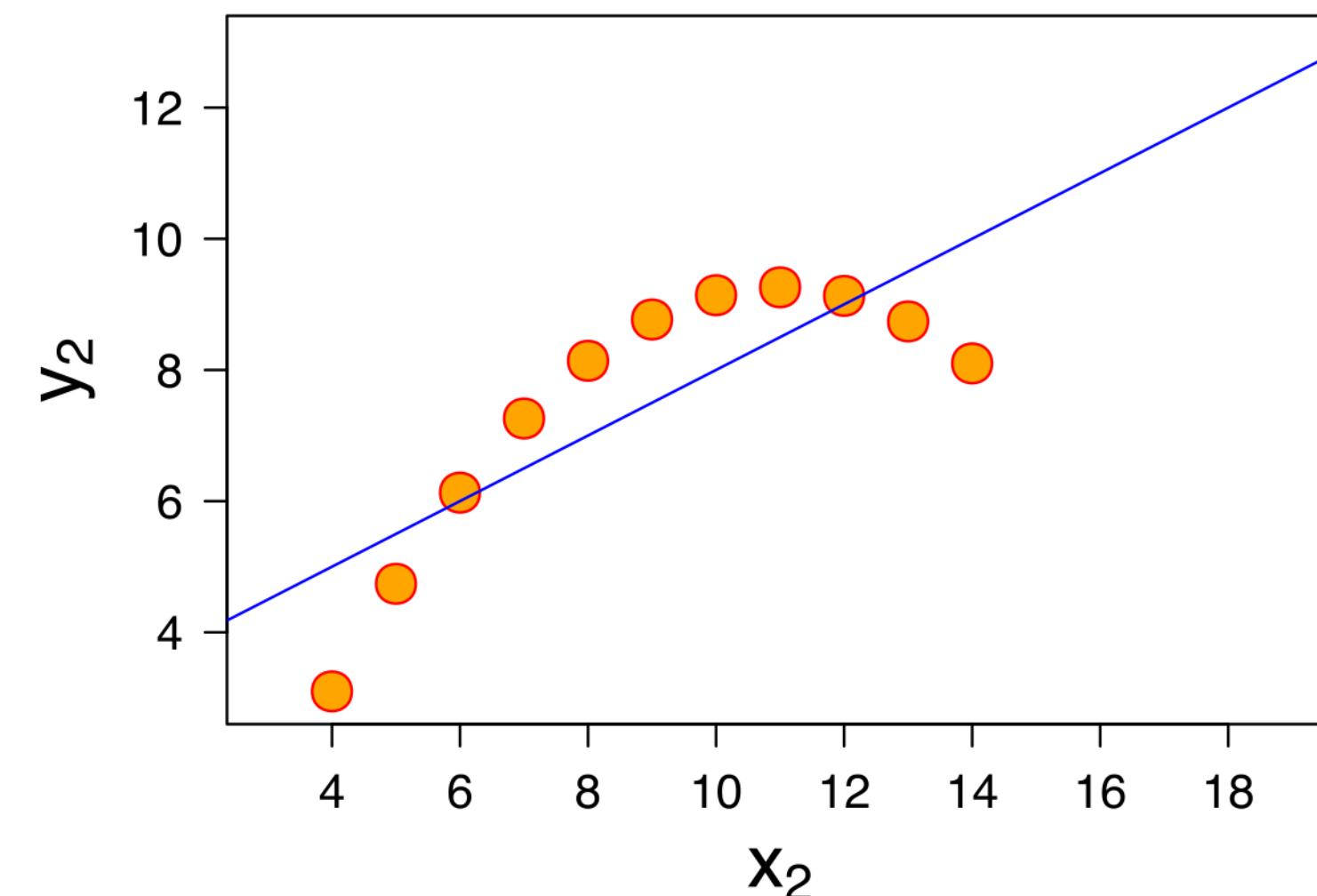
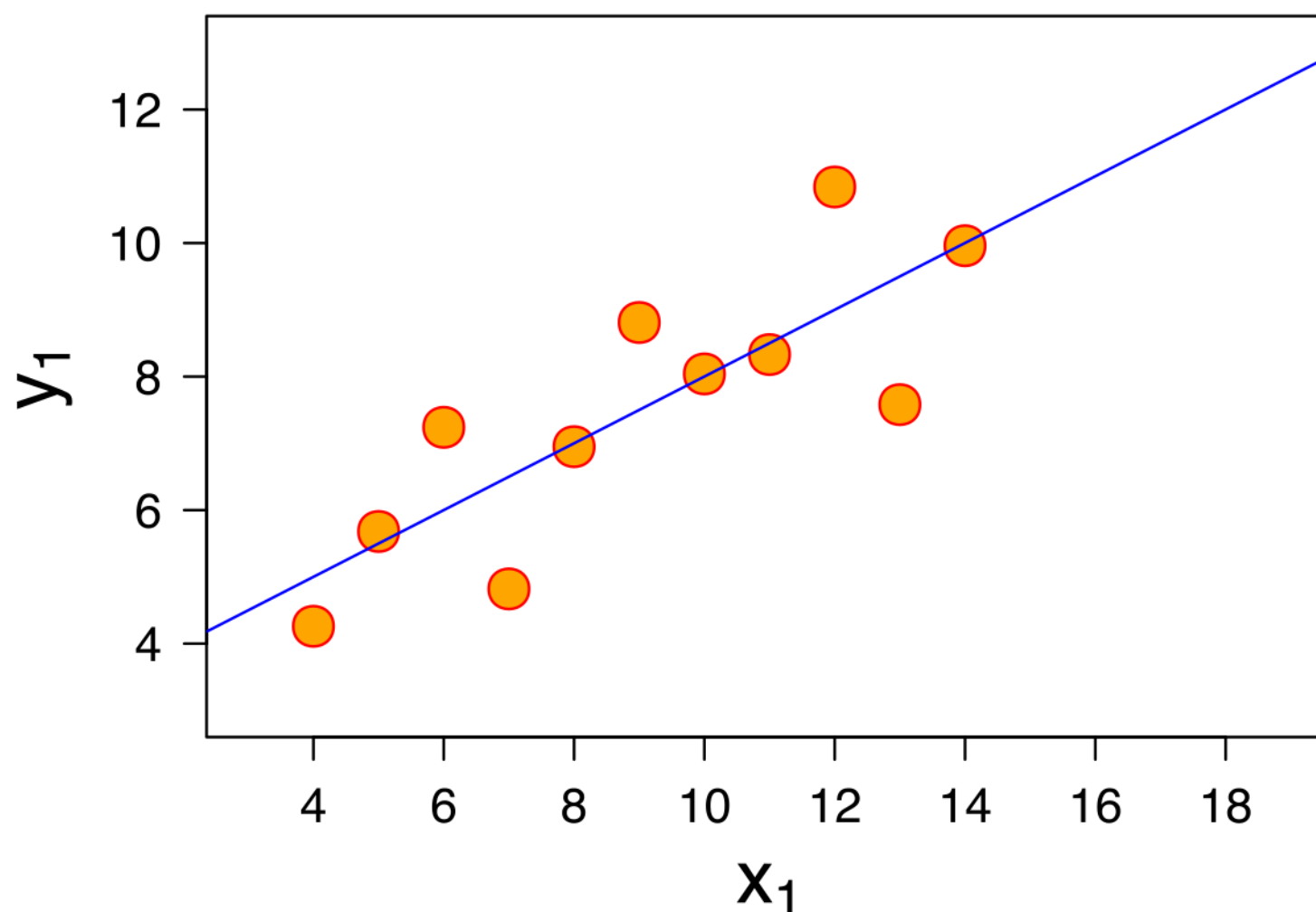
- Look up in table, or
- use `scipy.stats.chi2.sf()`

Agreement?

- General guidance:
 - Comparing Experimental Results with an accepted answer (Hughes & Hase. 3.3.4, p.28):
 - up to one standard error, they are in excellent agreement ;
 - between one and two standard errors, they are in reasonable agreement ;
 - more than three standard errors, they are in disagreement .
- χ^2 test for goodness of fit (Hughes & Hase. 8.4, p.106):
 - very good agreement: $P(\chi_{min}^2; \nu) \sim 0.5$
 - If $P(\chi_{min}^2; \nu) \rightarrow 1$, uncertainties are too large or data is too perfect
 - The agreement is questionable if $P(\chi_{min}^2; \nu) \approx 10^{-3}$.
 - The agreement is bad if $P(\chi_{min}^2; \nu) \lesssim 10^{-4}$

Always visually inspect your data:

- Visual inspection can reveal trends to the eye that are not picked up by statistics! (especially outliers!)
- Example, Anscombe's quartet:
 - all four data sets have the same mean and variance for x & y , and the same linear fit parameters and coefficient of correlation





A Complete Example

- https://github.com/UCD-Physics/Python-HowTos/blob/main/Curve_fit.ipynb